

R2P Program Unit 4 - Lesson 1

Quantum Diagonalization Methods

Sean Wagner, Ph.D.
Research Scientist | Quantum Technical Ambassador
IBM Quantum

Overview

Introduction

- Objective
- Current approaches and their limitations

Quantum-Centric Supercomputing

- Motivation
- Current state

Sample-Based methods

- Sample-based Quantum Diagonalization (SQD)
- Krylov Quantum Diagonalization (KQD)
- Sample-based Krylov Quantum Diagonalization (SKQD)

Introduction

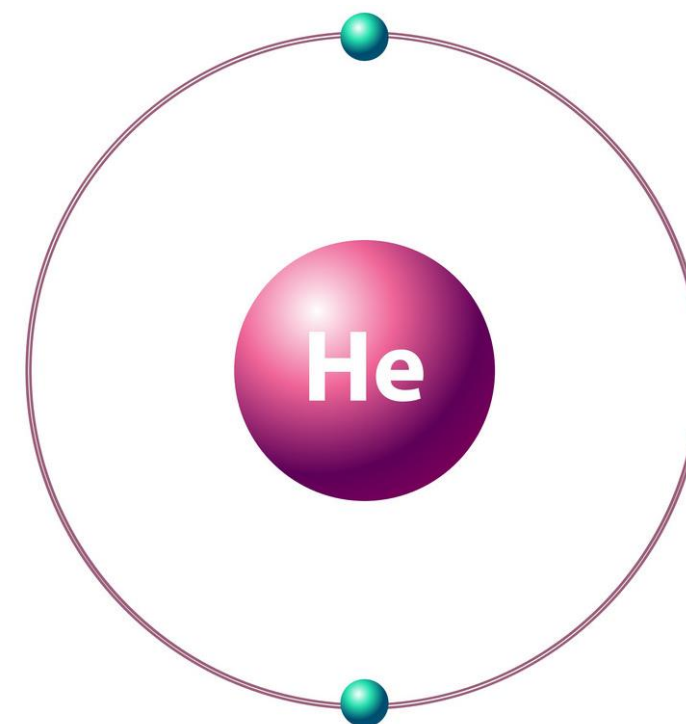
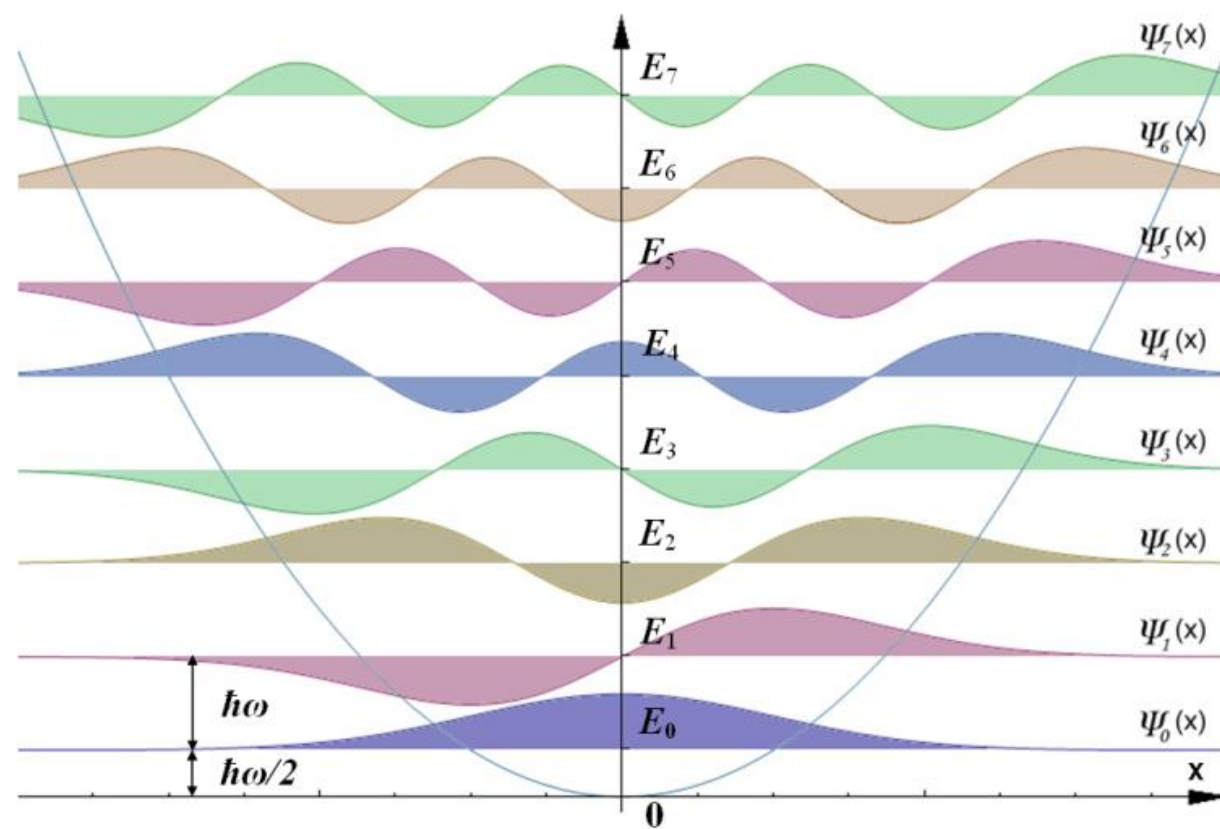
Searching the ground state

Goal: finding approaches that solves efficiently the following equation:

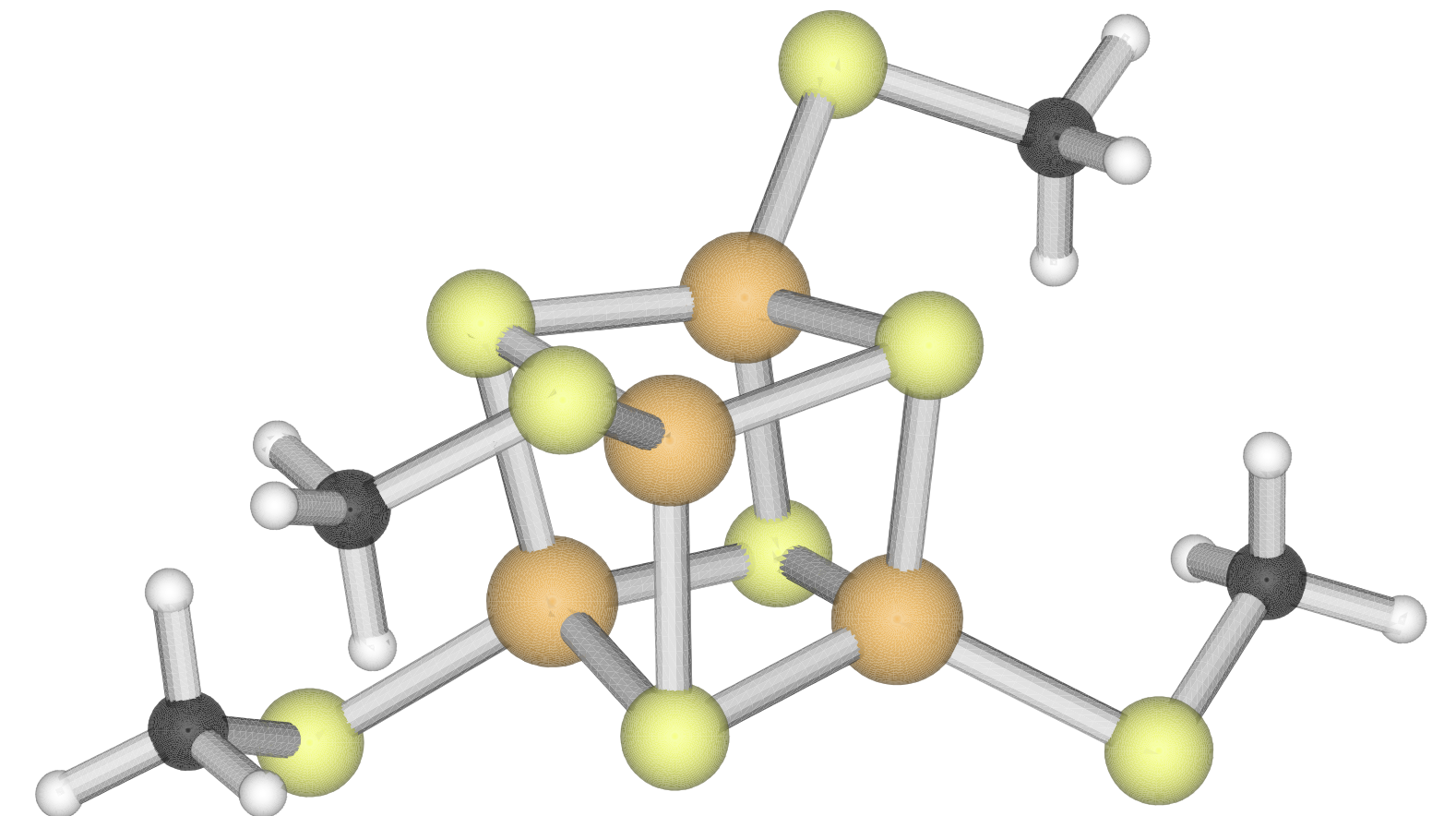
$$H|\psi\rangle = E|\psi\rangle$$

Why important? Looking for ground state and excited states energy is fundamental goal in chemistry, material science, physics

Challenge: As the system grows, the Hilbert space exponentially grows → Hard to diagonalize



$$H = -\frac{\hbar^2}{2m} (\nabla_1^2 + \nabla_2^2) - \frac{e^2}{4\pi\epsilon_0} \left[\frac{2}{r_1} + \frac{2}{r_2} - \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} \right]$$



Complexity

Estimating ground state properties is relevant across disciplines

High-Energy Physics

Understanding the fundamental nature of particles and forces

Materials

Predicting and understanding material behavior

Optimization

Discovering of optimal solutions in complex landscapes

Healthcare & Life Sciences

Understanding biochemical interactions and reactions

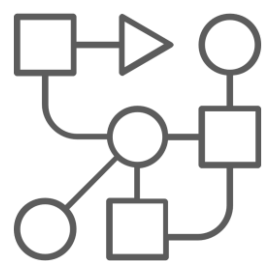
Current classical/quantum techniques to estimate ground state properties face challenges



Classical methods are hindered by the exponential growth of the Hilbert space with the system size and the correlations arising in entangled quantum systems.



Variational quantum methods (e. g. VQE) face challenges with current devices because of trainability and the high number of measurements required in the optimization process.



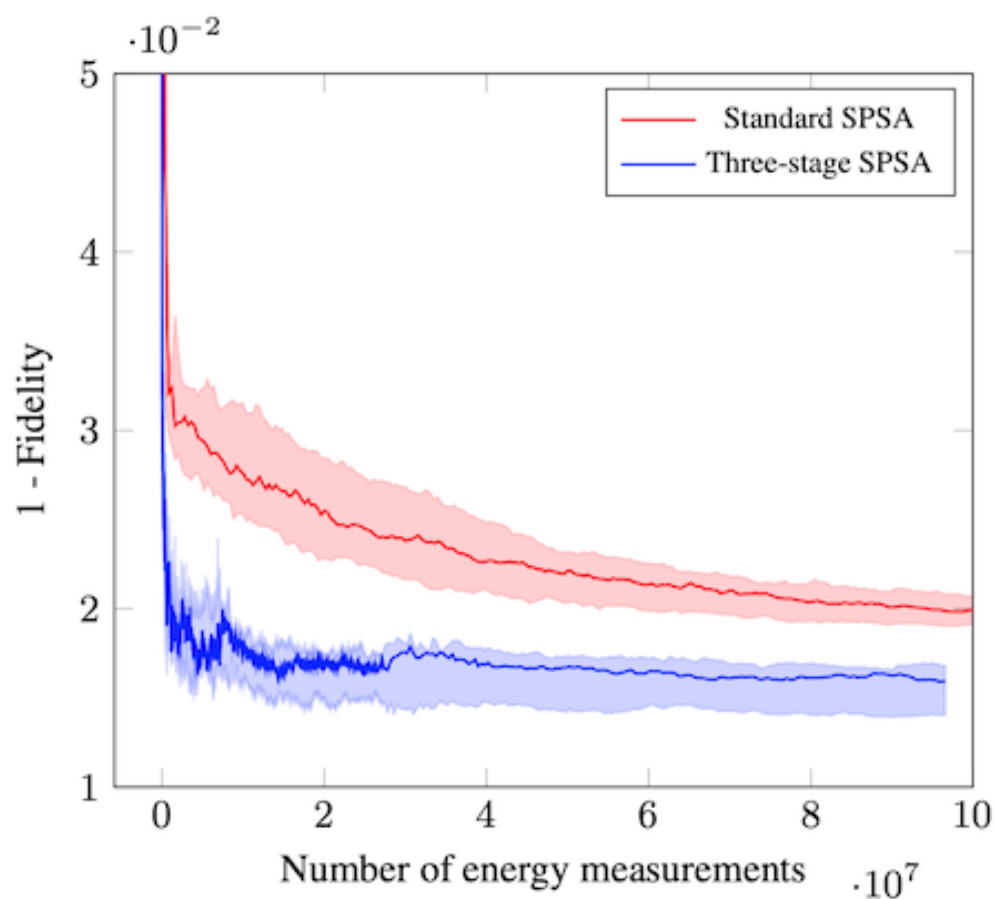
Fault-tolerant quantum methods with performance guarantees (e. g. quantum phase estimation) require deep circuits that cannot be executed on pre-fault-tolerant devices.

Limitations of VQE

Resource Scaling

The number of measurements and classical optimization steps grows quickly with system size.

- Measurement problem

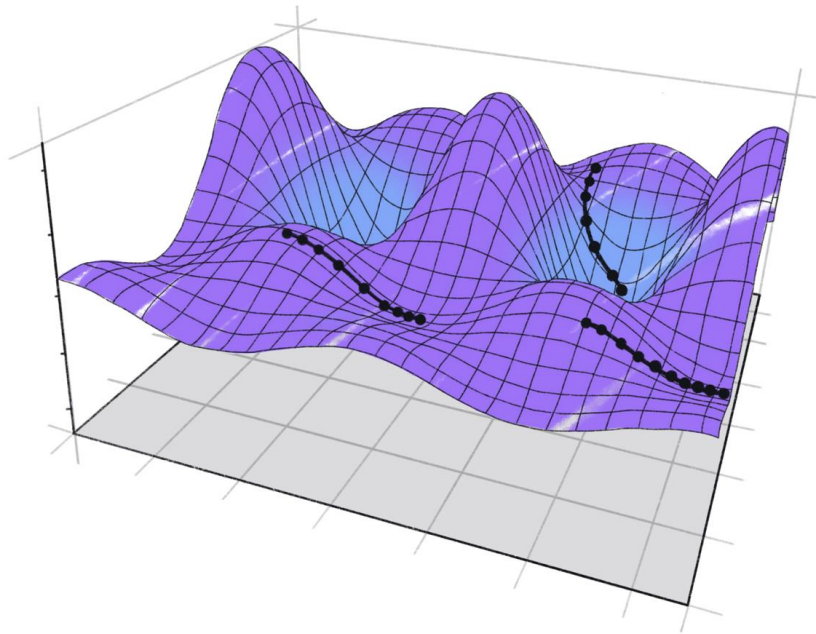
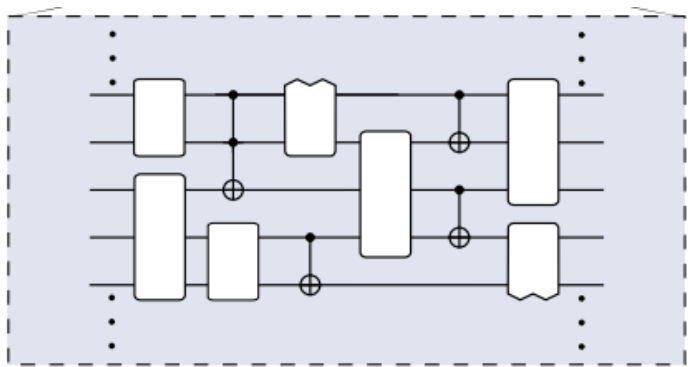


[C Cade *et al*, *PRB* **102**, 235122 (2020)]

Expressibility vs Trainability

The ansatz must be expressive enough to represent the ground state, but highly expressive ansatzes often lead to barren plateaus.

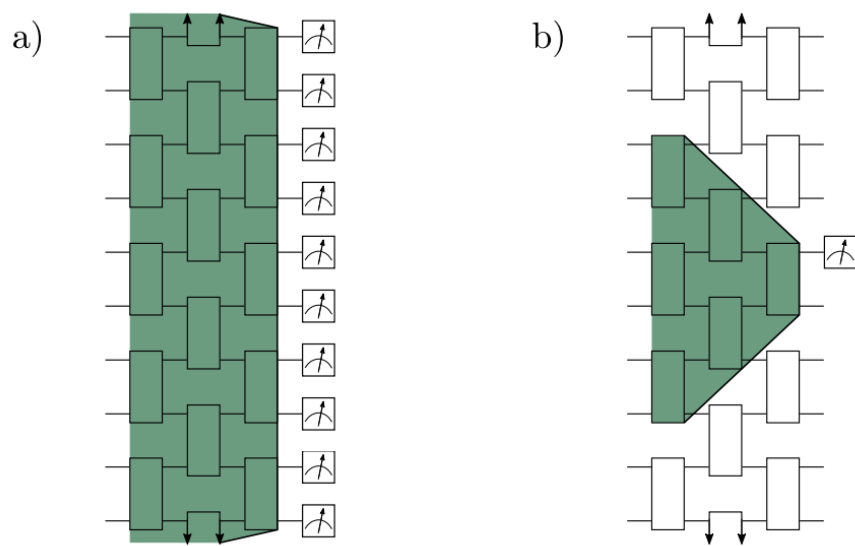
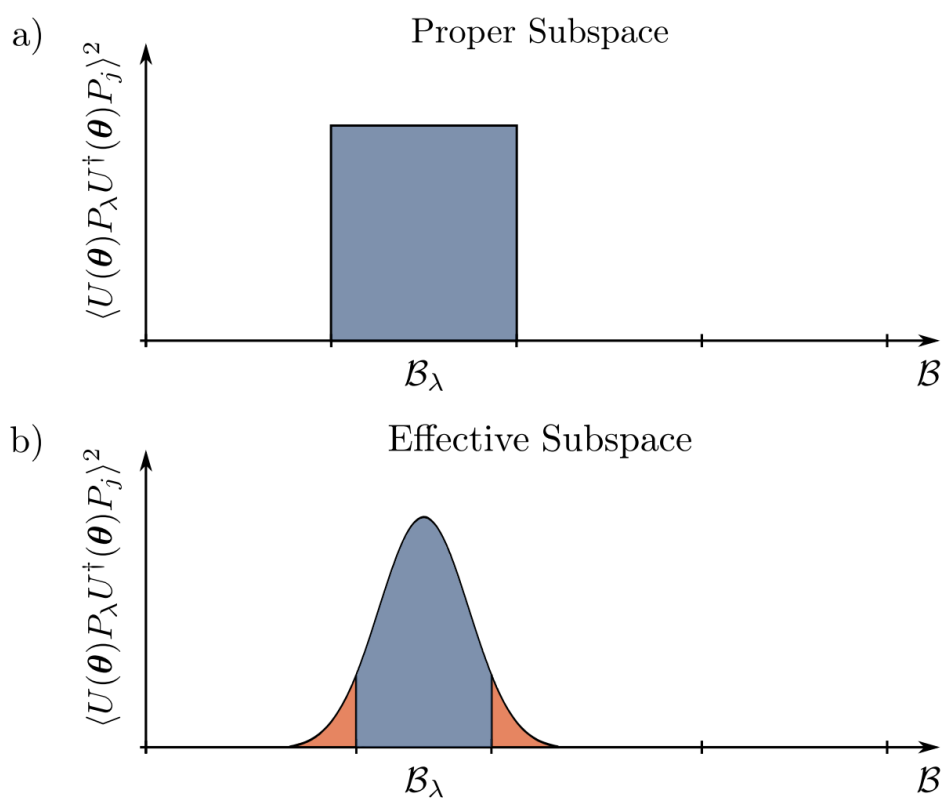
- Barren Plateaus
- Poor local minima



[Zimborás, Zoltán, *et al*. *arXiv*:2501.05694]

Classically Simulable

Many VQE circuits (especially shallow ones) can be efficiently simulated on classical computers.

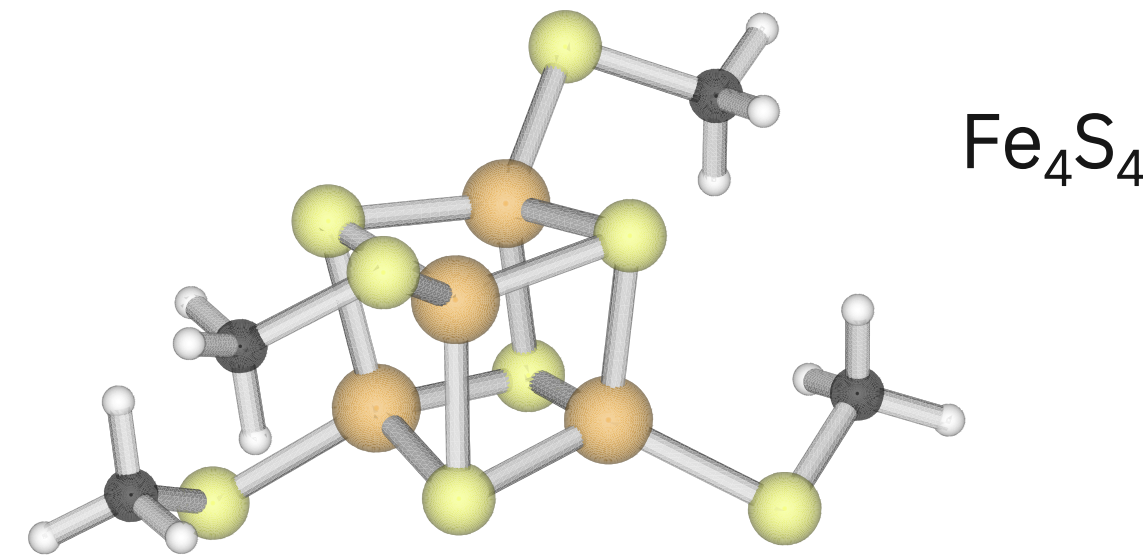


[M Cerezo *et al*, *arXiv*:2312.09121]

Quantum-Centric Supercomputing

The cost for the accurate prediction of nature

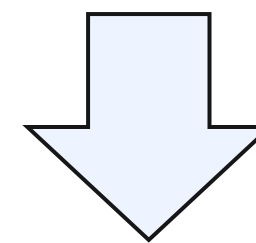
Goal: Compute ground-state energies by solving the Schrödinger equation $\hat{H}|\Psi\rangle = E|\Psi\rangle$ in the Born Oppenheimer approximation with given precision.



$$\hat{H} = \sum_{\substack{pr \\ \sigma}} h_{pr} \hat{a}_{p\sigma}^\dagger \hat{a}_{r\sigma} + \sum_{\substack{prqs \\ \sigma\tau}} \frac{(pr|qs)}{2} \hat{a}_{p\sigma}^\dagger \hat{a}_{q\tau}^\dagger \hat{a}_{s\tau} \hat{a}_{r\sigma}$$

- Fix atom nucleus
- Choose atomic basis set
- Solve Hartree Fock
- Write electron integrals in terms of the basis of molecular orbitals
- Map fermions to qubits

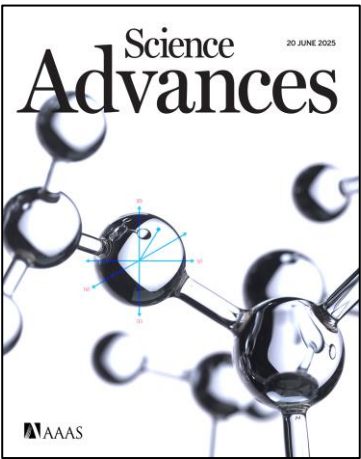
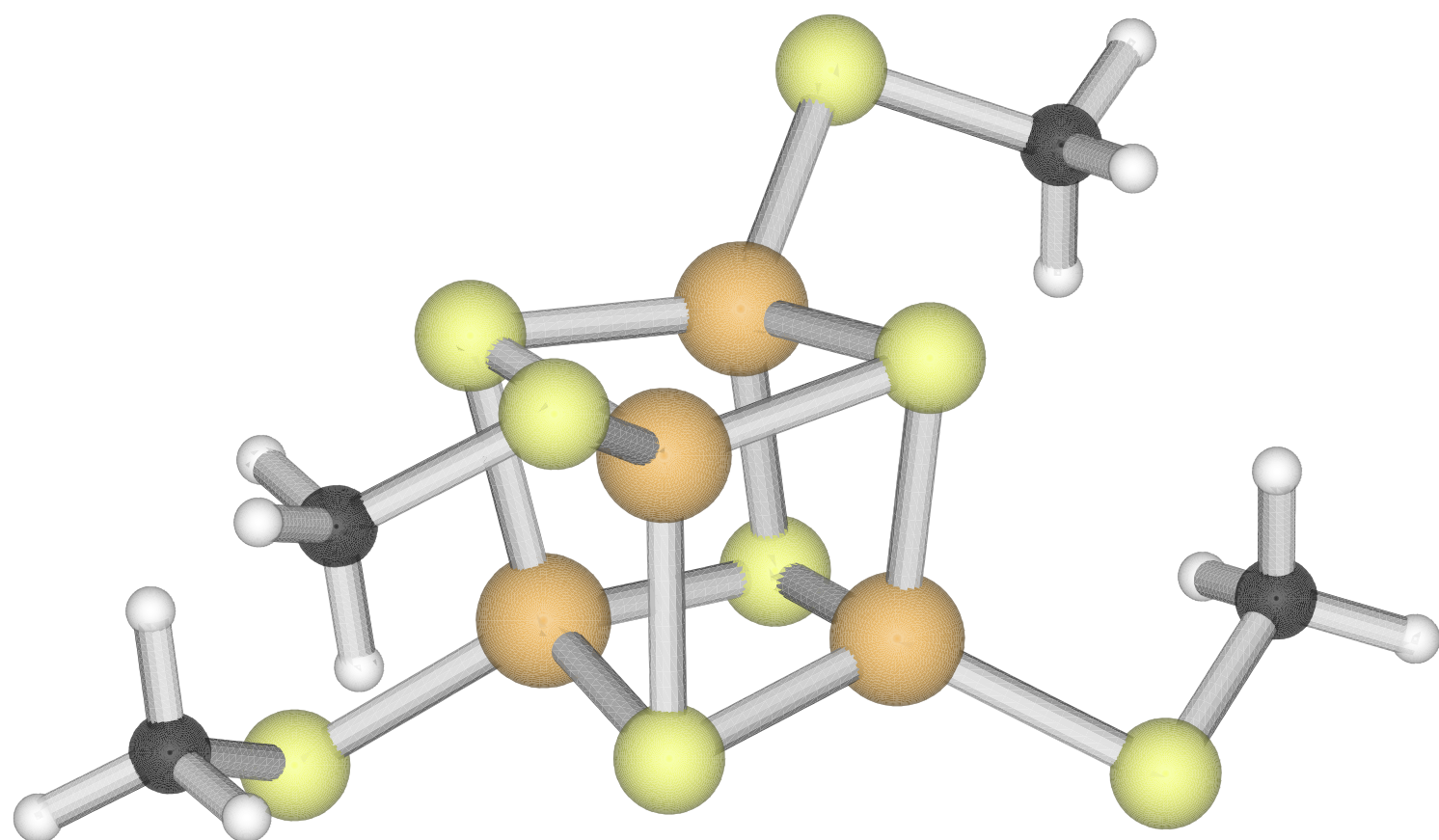
Fe₄S₄ on **72** qubits (TZP-DKH basis set): $\hat{H}$ has **6.7M** Pauli operators. To get a chemical (milliHartree) precision, on pre-fault tolerant quantum computers, each circuit must be executed **10¹⁴** times Runtime at 10μs/circuit ~**3M years**



What if we leverage also classical computers to obtain the expectation values of $\hat{H}$?

Quantum-centric supercomputing is unlocking new applications for near-term quantum hardware

Evaluating the ground-state energy for [4Fe-4S], a molecule **beyond exact diagonalization** scale.

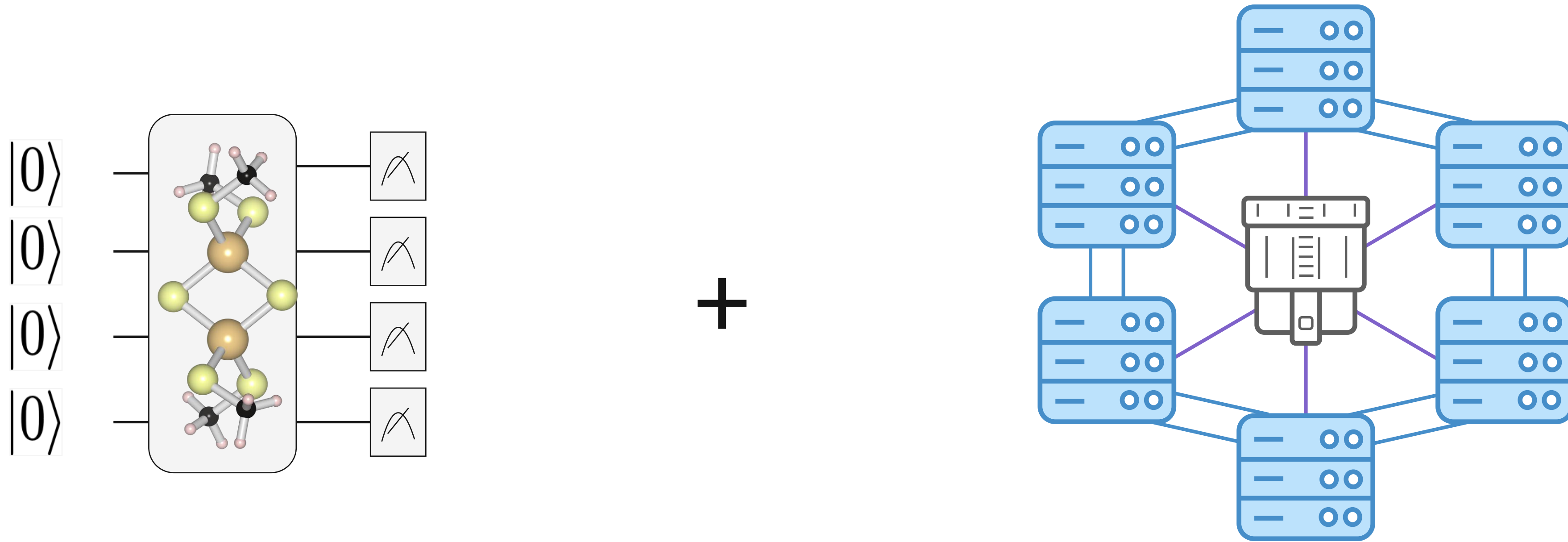


Science Advances vol. 11, no. 25 (2025)

Conventional Quantum	VQE estimation at 10μs/circuit ~3M years
Quantum-centric supercomputing	Subspace estimation at 10μs/circuit ~2 hours
Fault tolerant	Phase estimation qubits: 4.53M 13 days runtime*

*Estimation done using surface code

Chemistry on quantum-centric supercomputing



Sample from quantum circuits that accurately prepare the quantum probability distribution of a set of states

Use massive classical computing to process individual quantum samples and obtain an expectation value

IBM and RIKEN: Quantum system 2 and supercomputer Fugaku

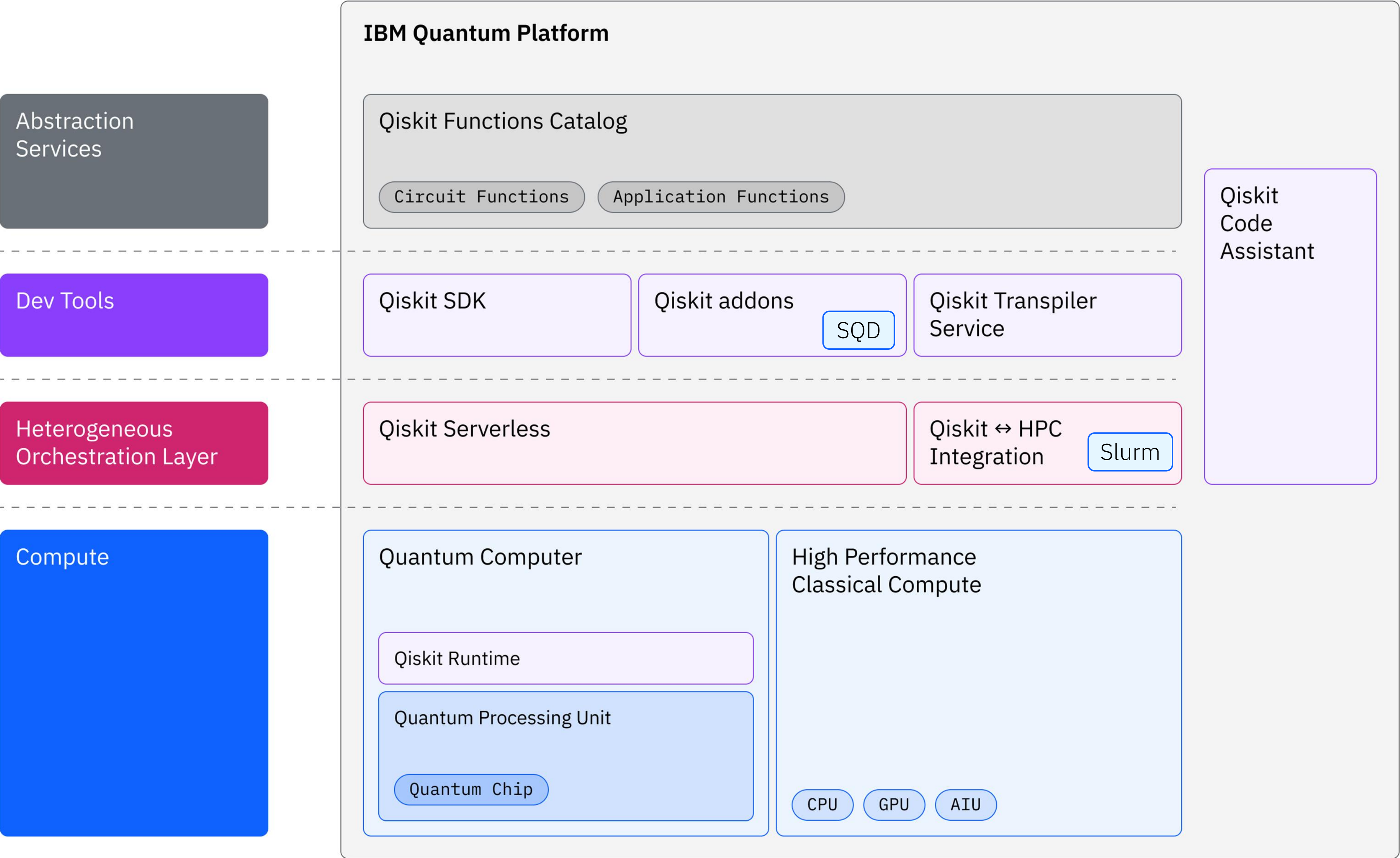
"By combining Fugaku and the IBM Quantum System Two, RIKEN aims to lead Japan into a new era of high-performance computing.

Our mission is to develop and demonstrate practical quantum-HPC hybrid workflows that can be explored by both the scientific community and industry. The connection of these two systems enables us to take critical steps toward realizing this vision."

Dr. Mitsuhsa Sato,
Division Director of the Quantum-HPC Hybrid
Platform Division,
RIKEN Center for Computational Science



QCSC Full stack



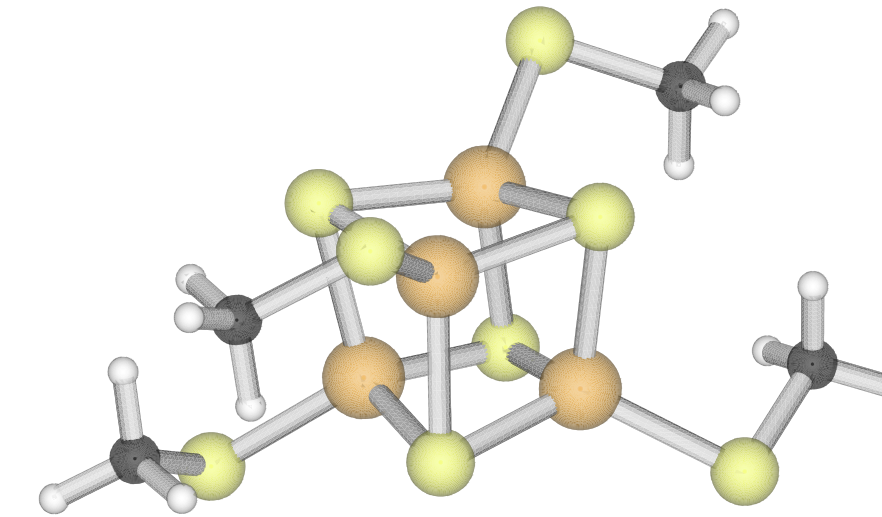
Quantum Diagonalization Methods

Quantum Diagonalization Methods

1. Sample-based Quantum Diagonalization (SQD) May 2024

Chemistry Beyond Exact Solutions on a Quantum-Centric Supercomputer

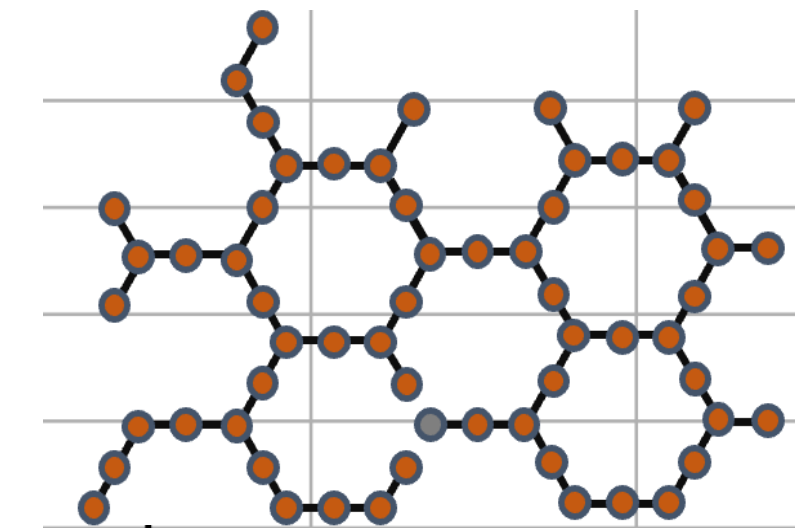
Javier Robledo-Moreno,^{1,*} Mario Motta,^{1,†} Holger Haas,¹ Ali Javadi-Abhari,¹ Petar Jurcevic,¹ William Kirby,² Simon Martiel,³ Kunal Sharma,¹ Sandeep Sharma,⁴ Tomonori Shirakawa,^{5,6,7} Iskandar Sitdikov,¹ Rong-Yang Sun,^{5,6,7} Kevin J. Sung,¹ Maika Takita,¹ Minh C. Tran,² Seiji Yunoki,^{5,6,7,8} and Antonio Mezzacapo^{1,‡}



2. Krylov Quantum Diagonalization (KQD) October 2024

Diagonalization of large many-body Hamiltonians on a quantum processor

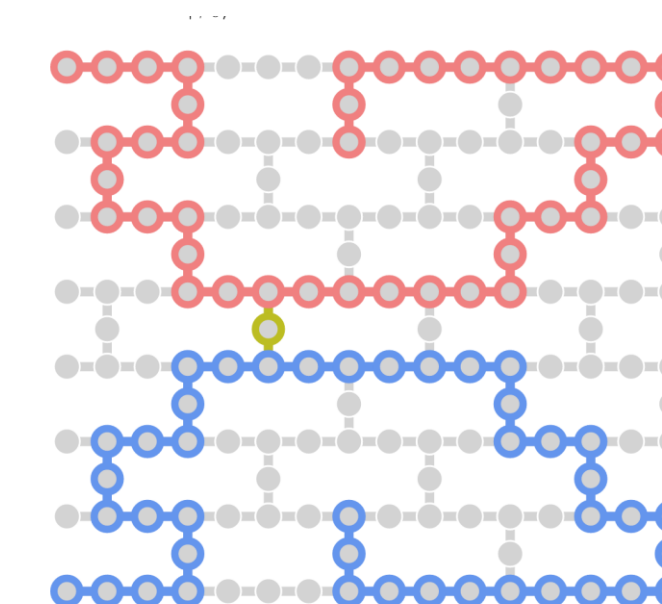
Nobuyuki Yoshioka*,^{1,†} Mirko Amico*,^{2,‡} William Kirby*,^{3,§} Petar Jurcevic,² Arkopal Dutt,³ Bryce Fuller,² Shelly Garion,⁴ Holger Haas,² Ikko Hamamura**,⁵ Alexander Ivrii,⁴ Ritajit Majumdar,⁶ Zlatko Minev,² Mario Motta,² Bibek Pokharel,⁷ Pedro Rivero,² Kunal Sharma,² Christopher J. Wood,² Ali Javadi-Abhari,² and Antonio Mezzacapo²



3. Sample-based Krylov Quantum Diagonalization (SKQD) January 2025

Sample-based Krylov Quantum Diagonalization

Jeffery Yu,^{1,2,3,*} Javier Robledo Moreno,^{1,†} Joseph Iosue,^{1,2,3} Luke Bertels,⁴ Daniel Claudino,⁴ Bryce Fuller,¹ Peter Groszkowski,⁵ Travis S Humble,⁶ Petar Jurcevic,¹ William Kirby,⁷ Thomas A. Maier,⁸ Mario Motta,¹ Bibek Pokharel,¹ Alireza Seif,¹ Amir Shehata,⁶ Kevin J. Sung,¹ Minh C. Tran,¹ Vinay Tripathi,¹ Antonio Mezzacapo,^{1,‡} and Kunal Sharma^{1,§}

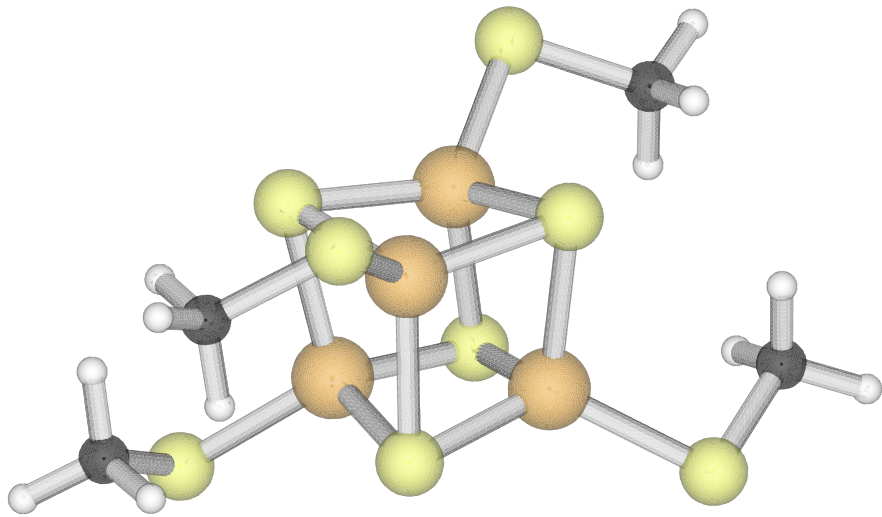


Quantum Diagonalization Methods

1. Sample-based Quantum Diagonalization (SQD) May 2024

Chemistry Beyond Exact Solutions on a Quantum-Centric Supercomputer

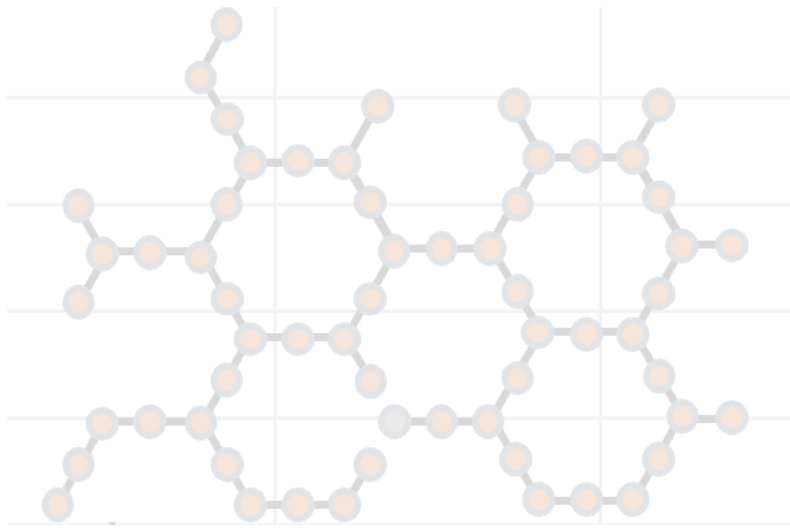
Javier Robledo-Moreno,^{1,*} Mario Motta,^{1,†} Holger Haas,¹ Ali Javadi-Abhari,¹ Petar Jurcevic,¹ William Kirby,² Simon Martiel,³ Kunal Sharma,¹ Sandeep Sharma,⁴ Tomonori Shirakawa,^{5,6,7} Iskandar Sitdikov,¹ Rong-Yang Sun,^{5,6,7} Kevin J. Sung,¹ Maika Takita,¹ Minh C. Tran,² Seiji Yunoki,^{5,6,7,8} and Antonio Mezzacapo^{1,‡}



2. Krylov Quantum Diagonalization (KQD) October 2024

Diagonalization of large many-body Hamiltonians on a quantum processor

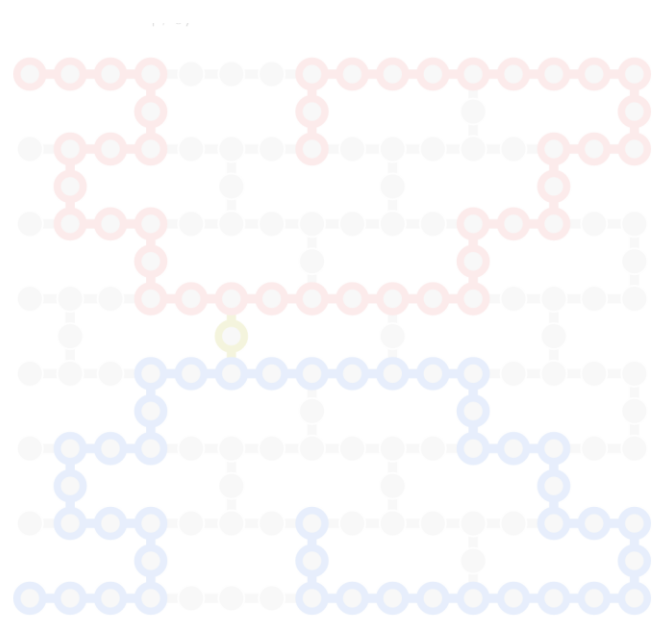
Nobuyuki Yoshioka*,^{1,†} Mirko Amico*,^{2,‡} William Kirby*,^{3,§} Petar Jurcevic,² Arkopal Dutt,³ Bryce Fuller,² Shelly Garion,⁴ Holger Haas,² Ikko Hamamura**,⁵ Alexander Ivrii,⁴ Ritajit Majumdar,⁶ Zlatko Minev,² Mario Motta,² Bibek Pokharel,⁷ Pedro Rivero,² Kunal Sharma,² Christopher J. Wood,² Ali Javadi-Abhari,² and Antonio Mezzacapo²



3. Sample-based Krylov Quantum Diagonalization (SKQD) January 2025

Sample-based Krylov Quantum Diagonalization

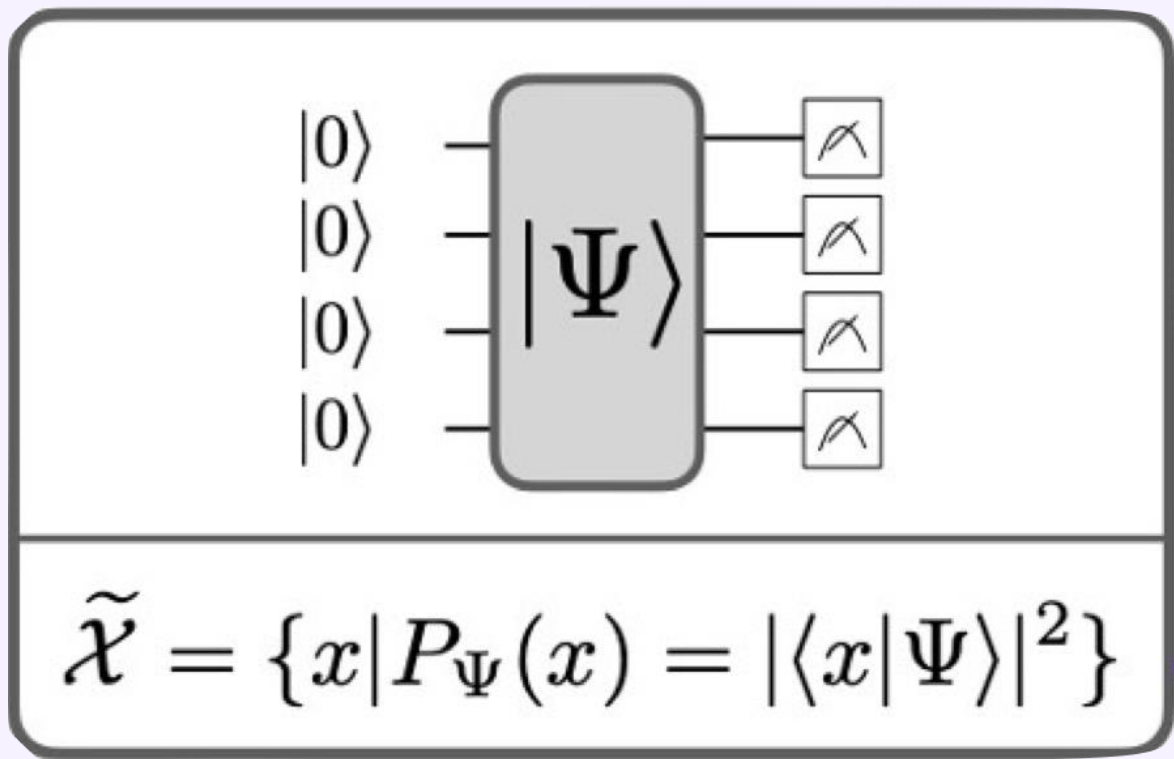
Jeffery Yu,^{1,2,3,*} Javier Robledo Moreno,^{1,†} Joseph Iosue,^{1,2,3} Luke Bertels,⁴ Daniel Claudino,⁴ Bryce Fuller,¹ Peter Groszkowski,⁵ Travis S Humble,⁶ Petar Jurcevic,¹ William Kirby,⁷ Thomas A. Maier,⁸ Mario Motta,¹ Bibek Pokharel,¹ Alireza Seif,¹ Amir Shehata,⁶ Kevin J. Sung,¹ Minh C. Tran,¹ Vinay Tripathi,¹ Antonio Mezzacapo,^{1,‡} and Kunal Sharma^{1,§}



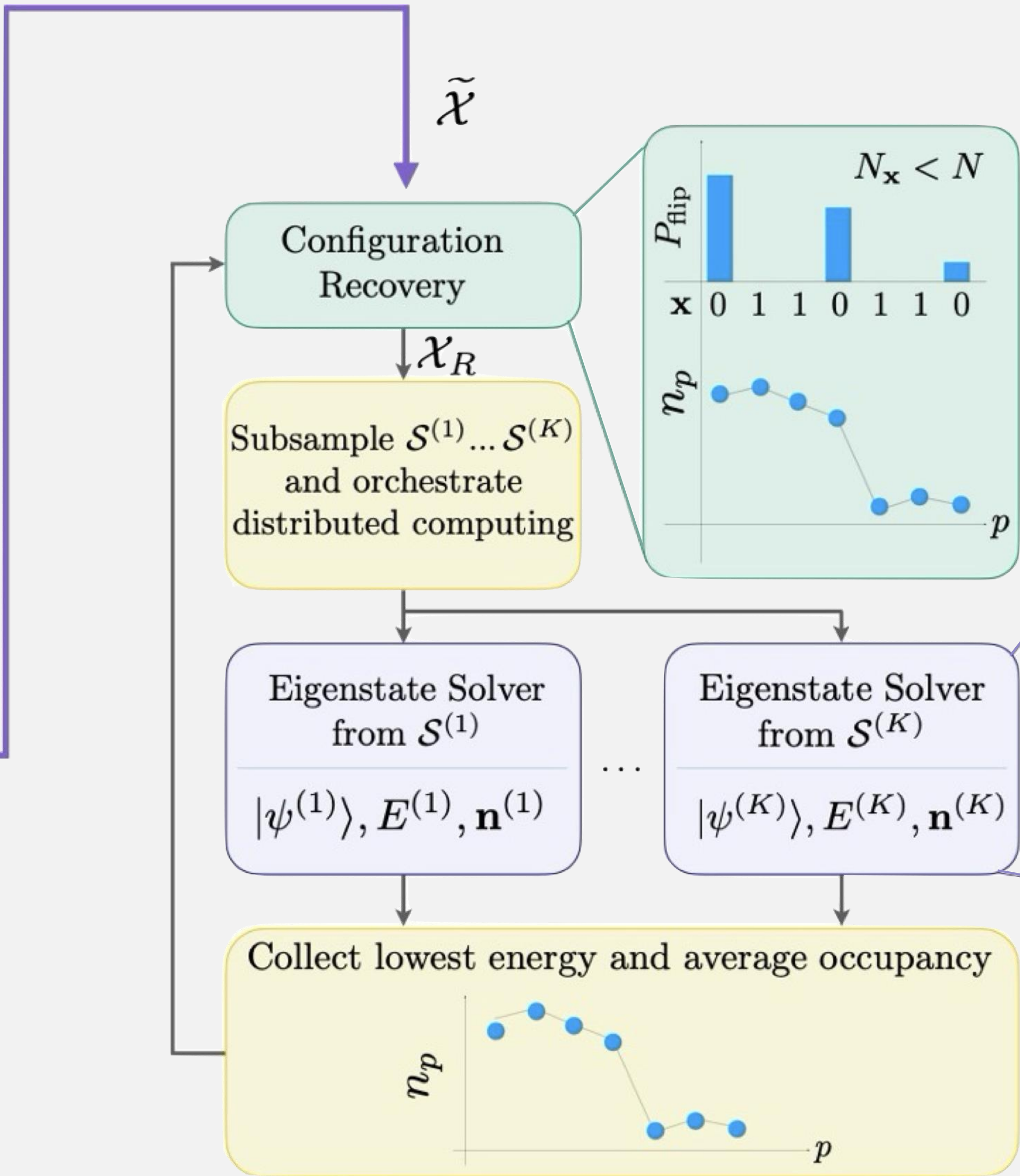
Sample-based Quantum Diagonalization (SQD)

Execute using Qiskit Runtime Primitives. Quantum compute generates samples from an exponentially large space

Sampler()



Efficient data transfer from QPU to HPC.



Post-process using potential advanced classical compute to extract meaningful information

$$H_{\mathcal{S}^{(k)}} = P_{\mathcal{S}^{(k)}} H P_{\mathcal{S}^{(k)}}$$

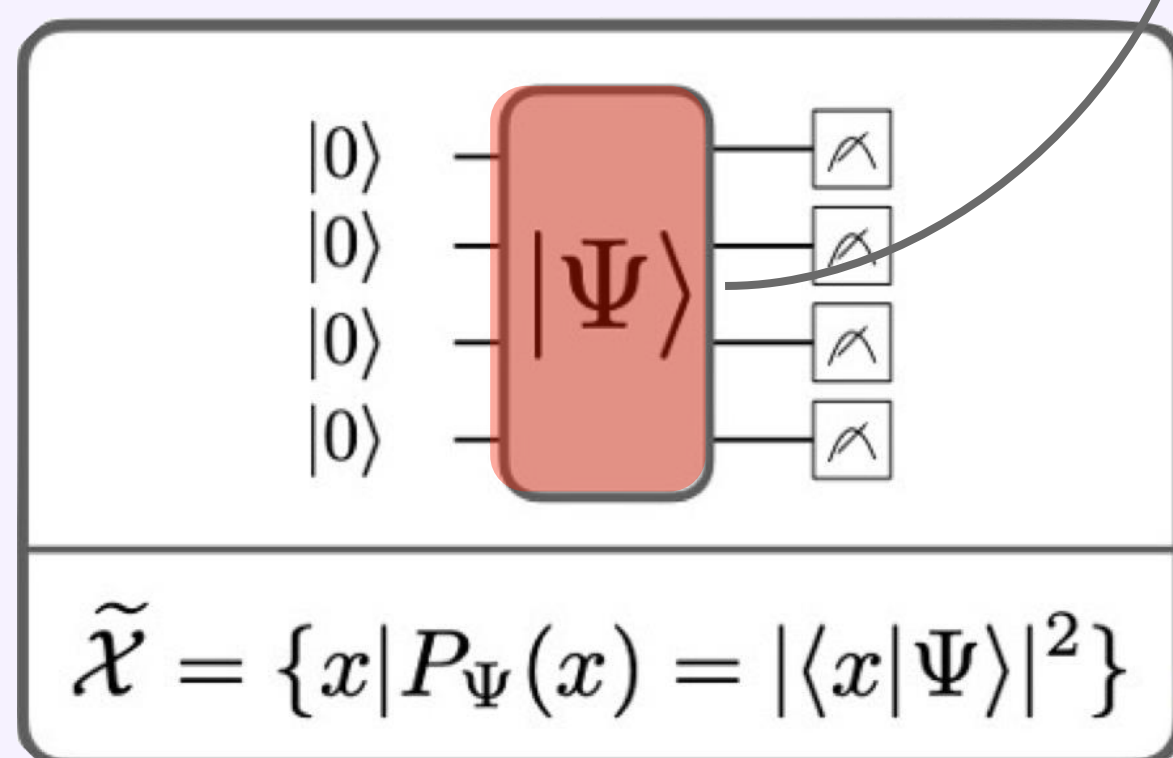
with $P_{\mathcal{S}^{(k)}} = \sum_{\mathbf{x} \in \mathcal{S}^{(k)}} |\mathbf{x}\rangle \langle \mathbf{x}|$

$$H_{\mathcal{S}^{(k)}} |\psi^{(k)}\rangle = E^{(k)} |\psi^{(k)}\rangle$$

Harrison, J. Chem. Phys. 94, 5021-5031
Kanno et al., arXiv 2302.11320

Sample-based Quantum Diagonalization (SQD)

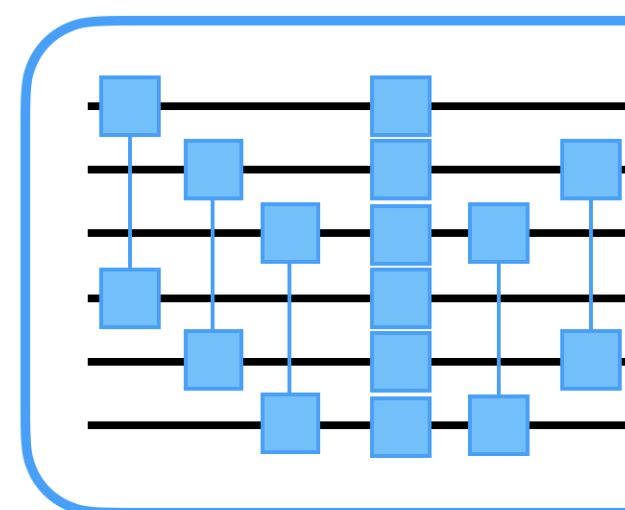
Sampler()



Step 1: prepare quantum state on QPU

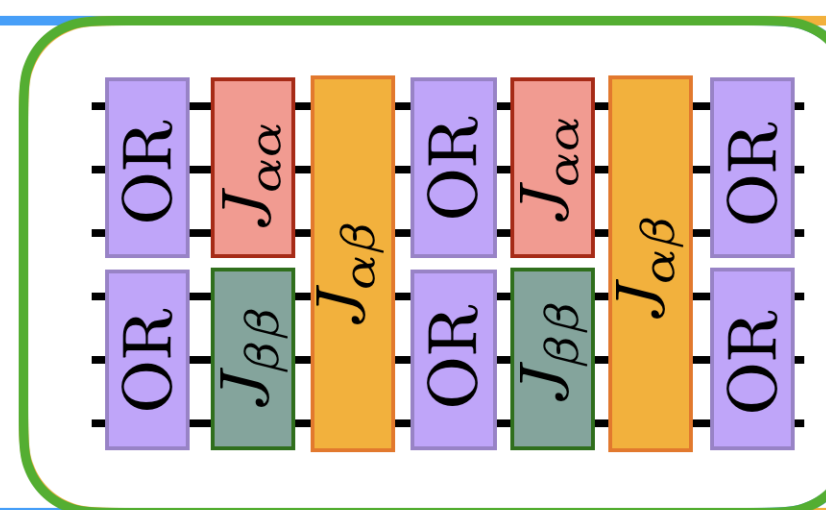
Prepare a quantum state $|\Psi\rangle$ on M qubits, which represents a molecular wavefunction on M molecular spin-orbitals.

Hardware-efficient



Compatible with hardware connectivity/gates/depth...
Hard to initialize and optimize

LUCJ circuit

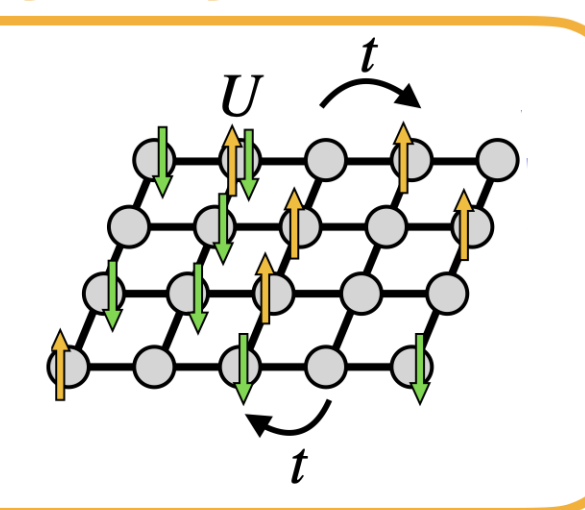


Local Unitary Cluster Jastrow

Physical motivation (derived from coupled-cluster)

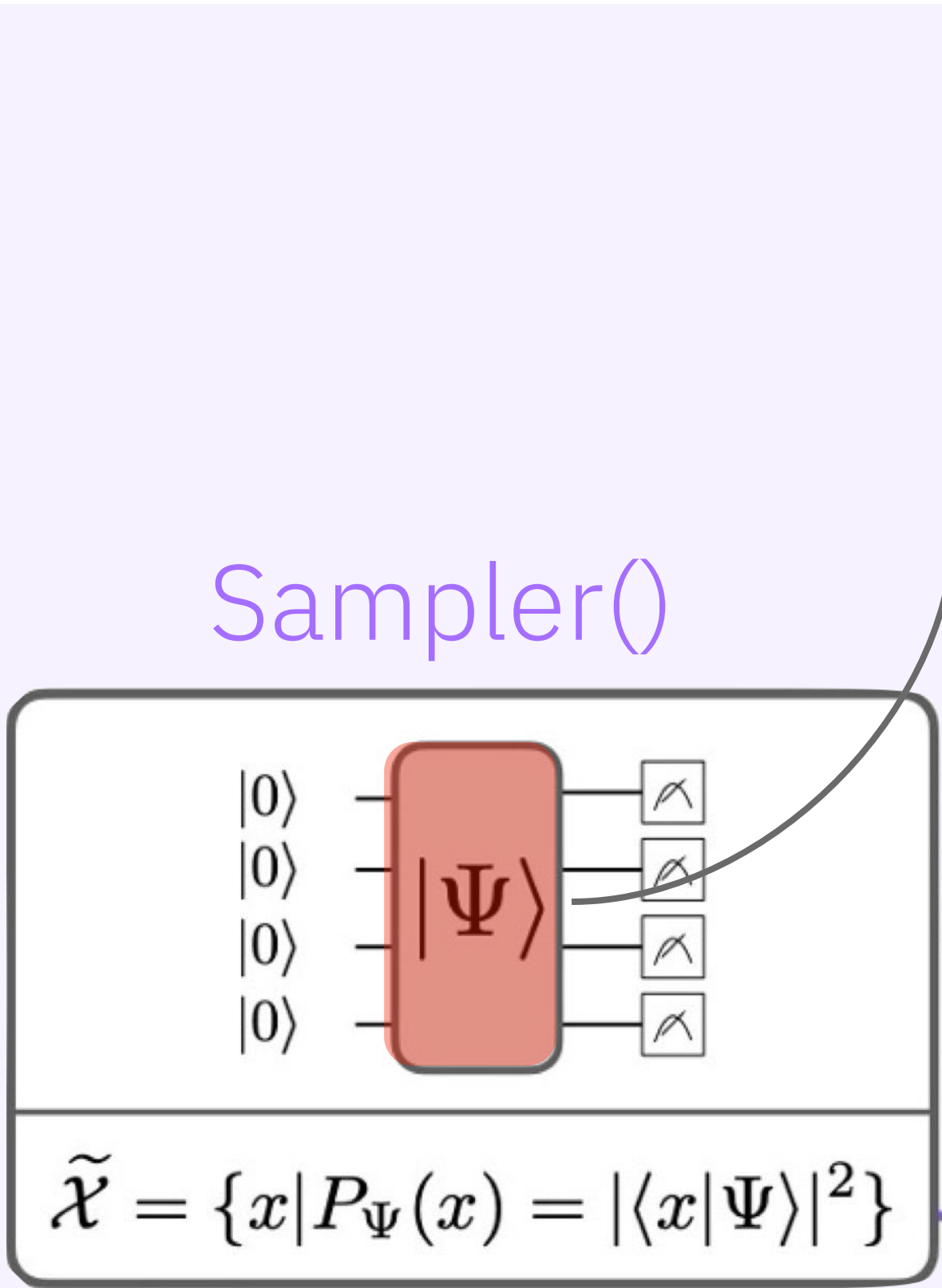
Hardware friendliness (device-specific connectivity)

Physically-motivated



Parameters from classical calculations, easy to optimize
All-to-all connectivity, high circuit depth and gate count

Sample-based Quantum Diagonalization (SQD)



Step 1: prepare quantum state on QPU

Unitary coupled-cluster singles and doubles (UCCSD)

$$|\Psi\rangle = e^{T-T^+} |\Phi_0\rangle$$
$$T = \sum_{ai} t_i^a \hat{c}_a^+ \hat{c}_i + \sum_{abij} t_{ij}^{ab} \hat{c}_a^+ \hat{c}_b^+ \hat{c}_j \hat{c}_i$$

Jastrow form of the UCCSD:

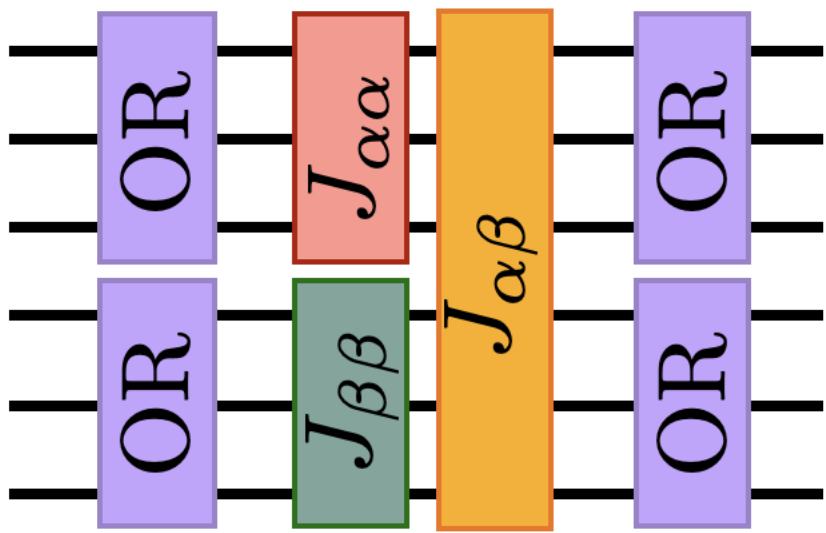
$$T - T^+ \simeq \sum_{\mu} e^{-K_{\mu}} iJ_{\mu} e^{K_{\mu}}$$

$$K_{\mu} = \sum_{pr,\sigma} K_{pr}^{\mu} \hat{c}_{p\sigma}^+ \hat{c}_{r\sigma}$$

$$J_{\mu} = \sum_{pr,\sigma\tau} J_{pr}^{\sigma\tau} \hat{n}_{p\sigma} \hat{n}_{r\tau}$$

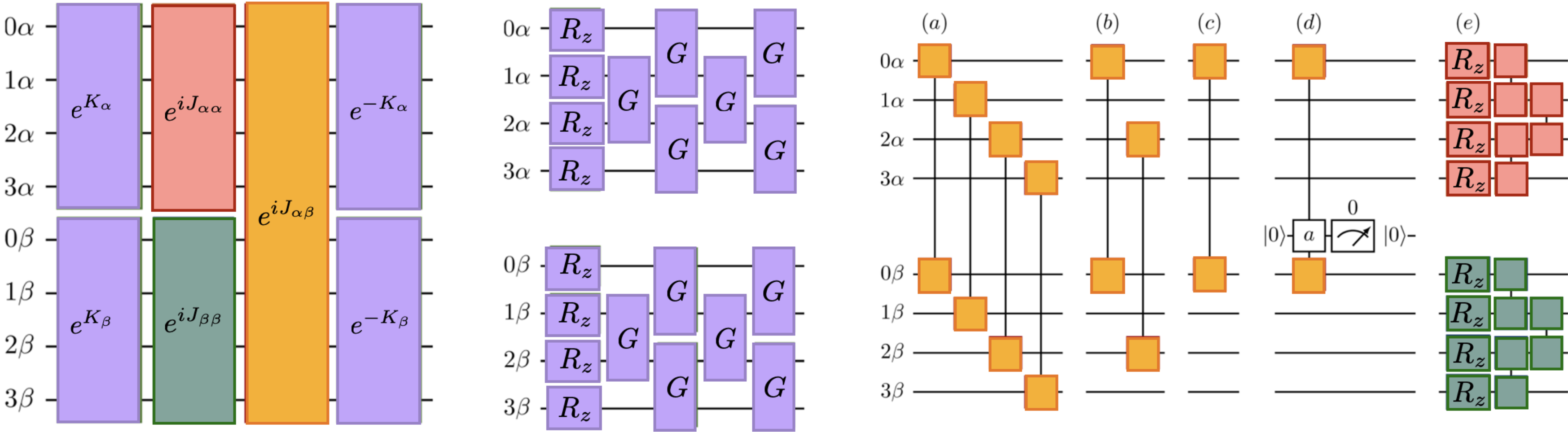
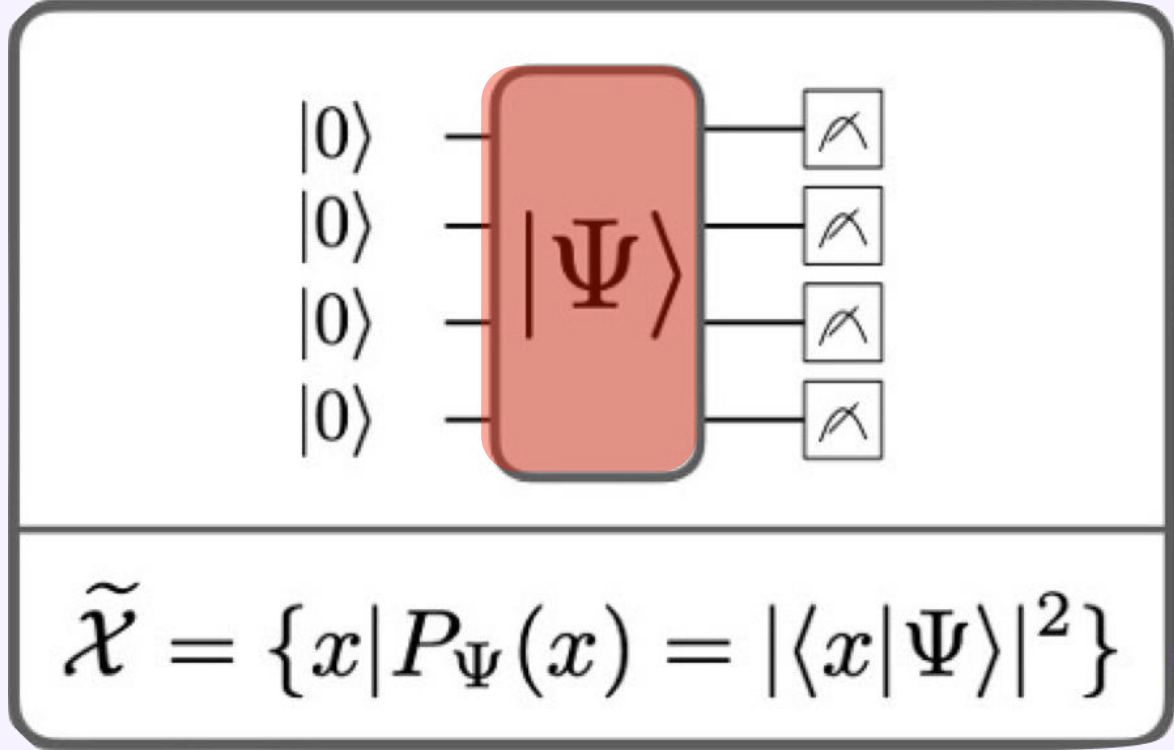
Local Unitary Cluster
Jastrow ansatz (LUCJ)

$$|\Psi\rangle \simeq \prod_{\mu} e^{-K_{\mu}} e^{iJ_{\mu}} e^{K_{\mu}} |\Phi_0\rangle$$



Sample-based Quantum Diagonalization (SQD)

Sampler()

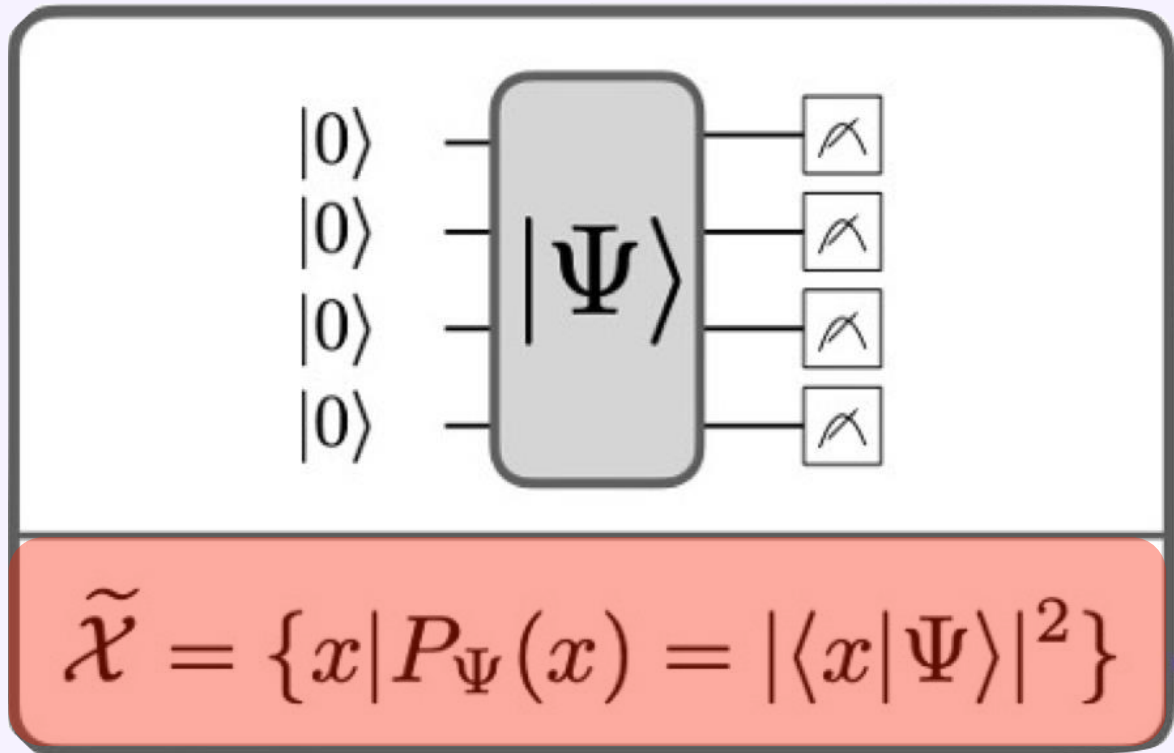


- These circuits can take as input parameters from classical solutions
- The circuits are hard to simulate classically*
- Compatible with quantum hardware

* Motta, Sung, Whaley, Head-Gordon, Shee, Chem. Sci. 14, 40, 11213-11227 (2023).

Sample-based Quantum Diagonalization (SQD)

Sampler()



Step 2: Sample from the quantum state

The circuit is then executed and measured in the computational basis.

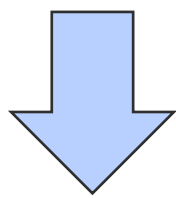
Repeating this produces a set of measurement outcomes

$$\tilde{\mathcal{X}} = \left\{ \mathbf{x} \mid \mathbf{x} \sim \tilde{P}_{\Psi}(\mathbf{x}) \right\}$$

1001, 1111
1000, 0111

The bitstrings $\mathbf{x} \in \{0, 1\}^M$ represent the different electronic configurations (Slater determinants).

The noise in the quantum circuit alters the distribution from its ideal form and generates the noisy set of configurations.



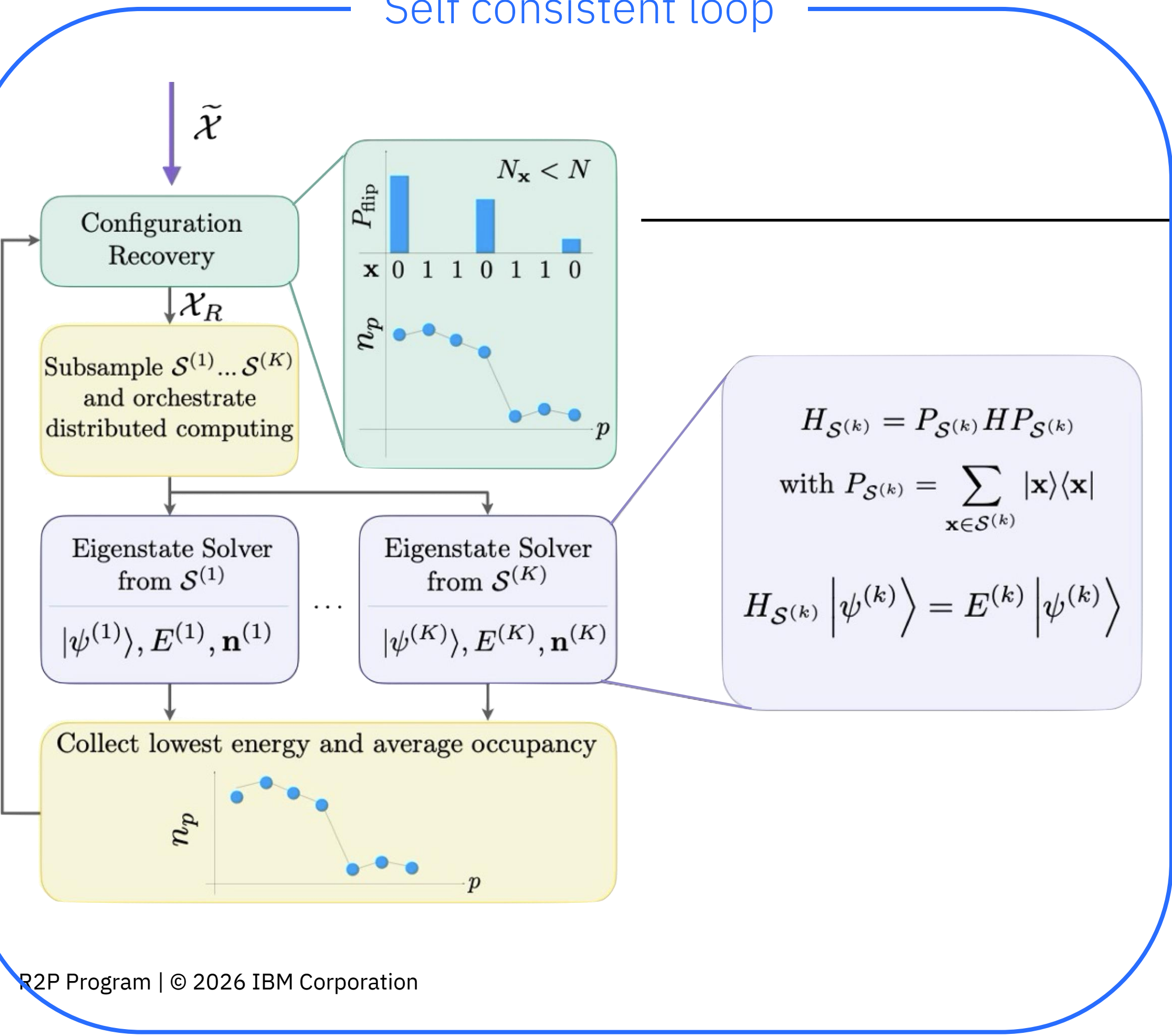
Probabilistic self-consistent Configuration Recovery

Sample-based Quantum Diagonalization (SQD)

Step 3: Setup + Configuration Recovery

uses information from collective properties of the wavefunction (particles number, etc) to refine noisy bitstrings into usable ones.

Self consistent loop



Setup: The initial guess for n used for the first round of the Configuration Recovery is calculated only from the initial bitstrings which belong in the correct particle sector (correct Hamming weight + correct population of the two spins up & down).

Configuration Recovery:

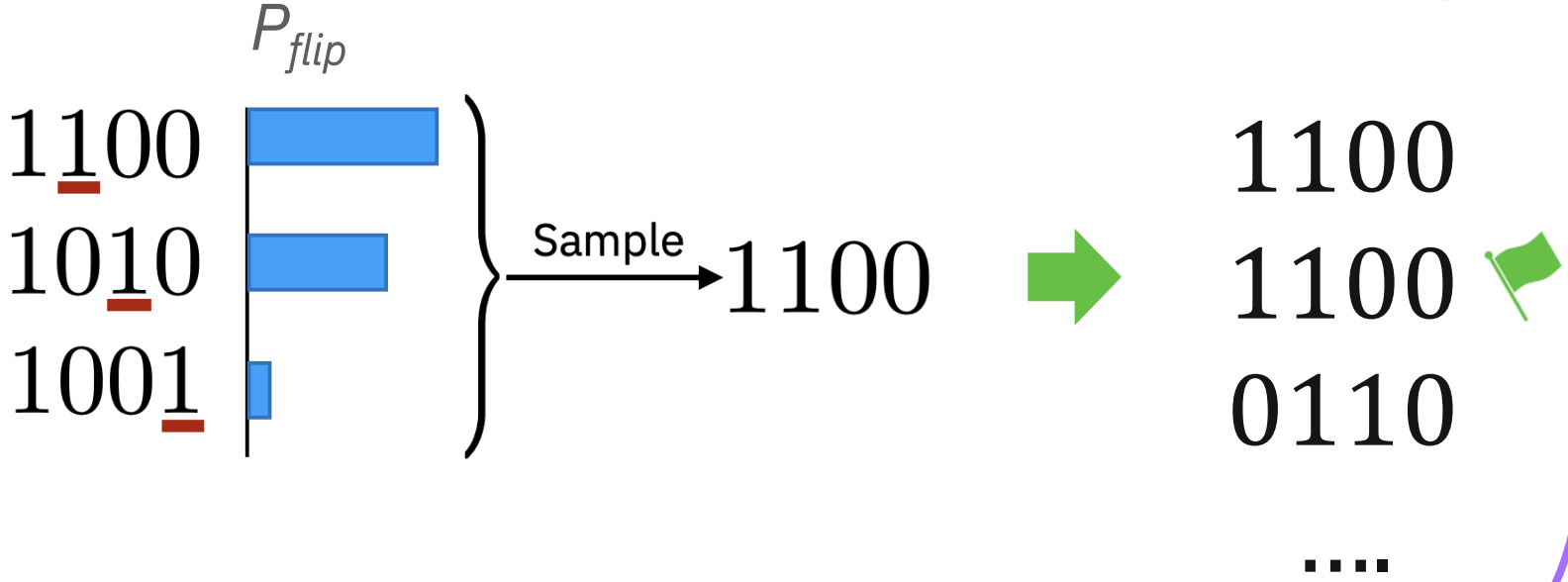
Bit strings after Setup:

1100
1000
0110
....



For $N=2$, 1000 is not representing electrons in the right particle sector.

Corrected bit string:

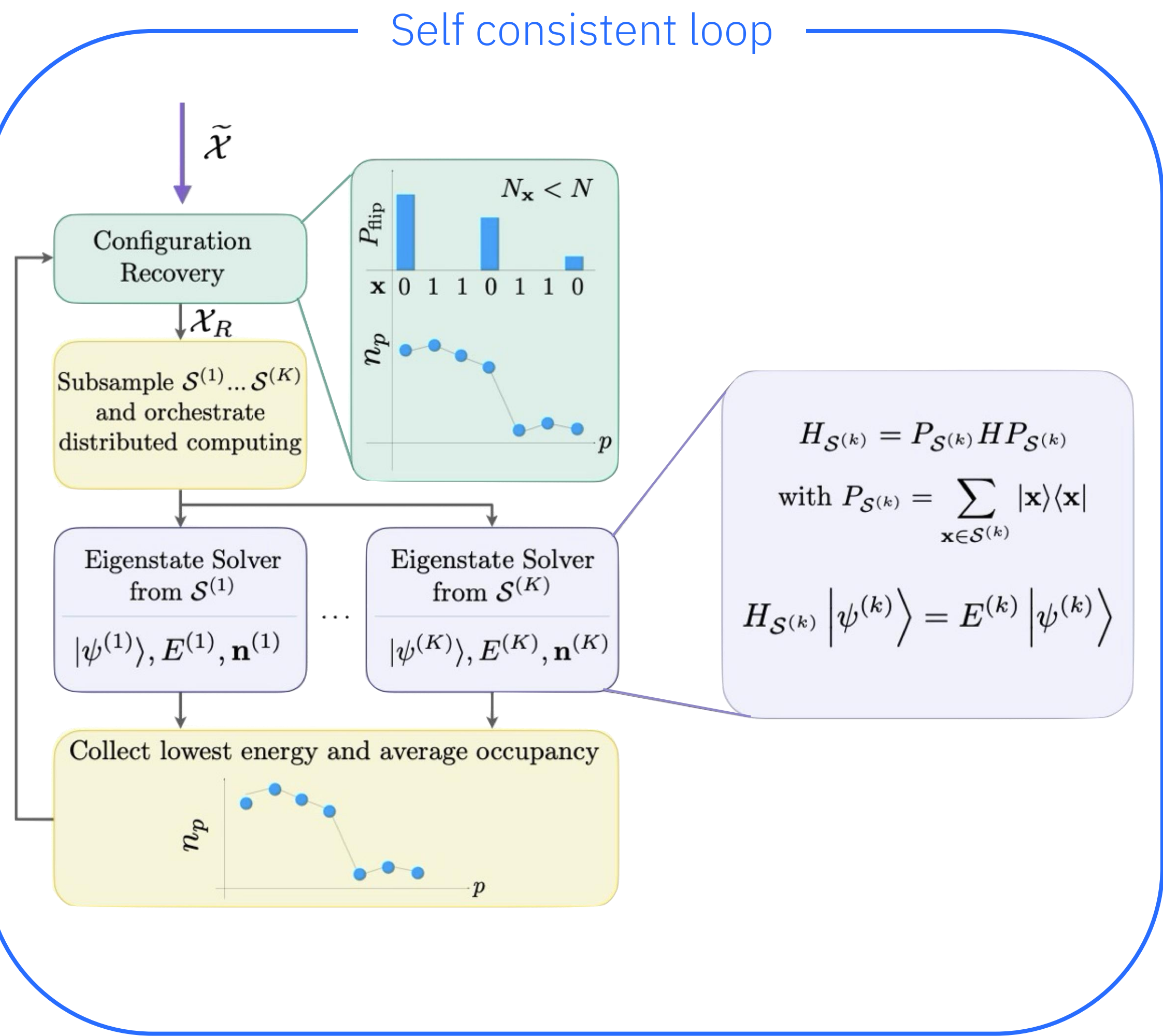


We correct the particles distribution in the correct particle sector by applying bit Flips according to P_{flip} depends on $|\mathbf{x}_{p\sigma} - n_{p\sigma}|$

Sample-based Quantum Diagonalization (SQD)

Step 4-5-6: Classical post-processing using an HPC

Self consistent loop



- Step 4

Create K batches of d configurations $\mathcal{S}(1) \dots, \mathcal{S}(K)$.
- Step 5

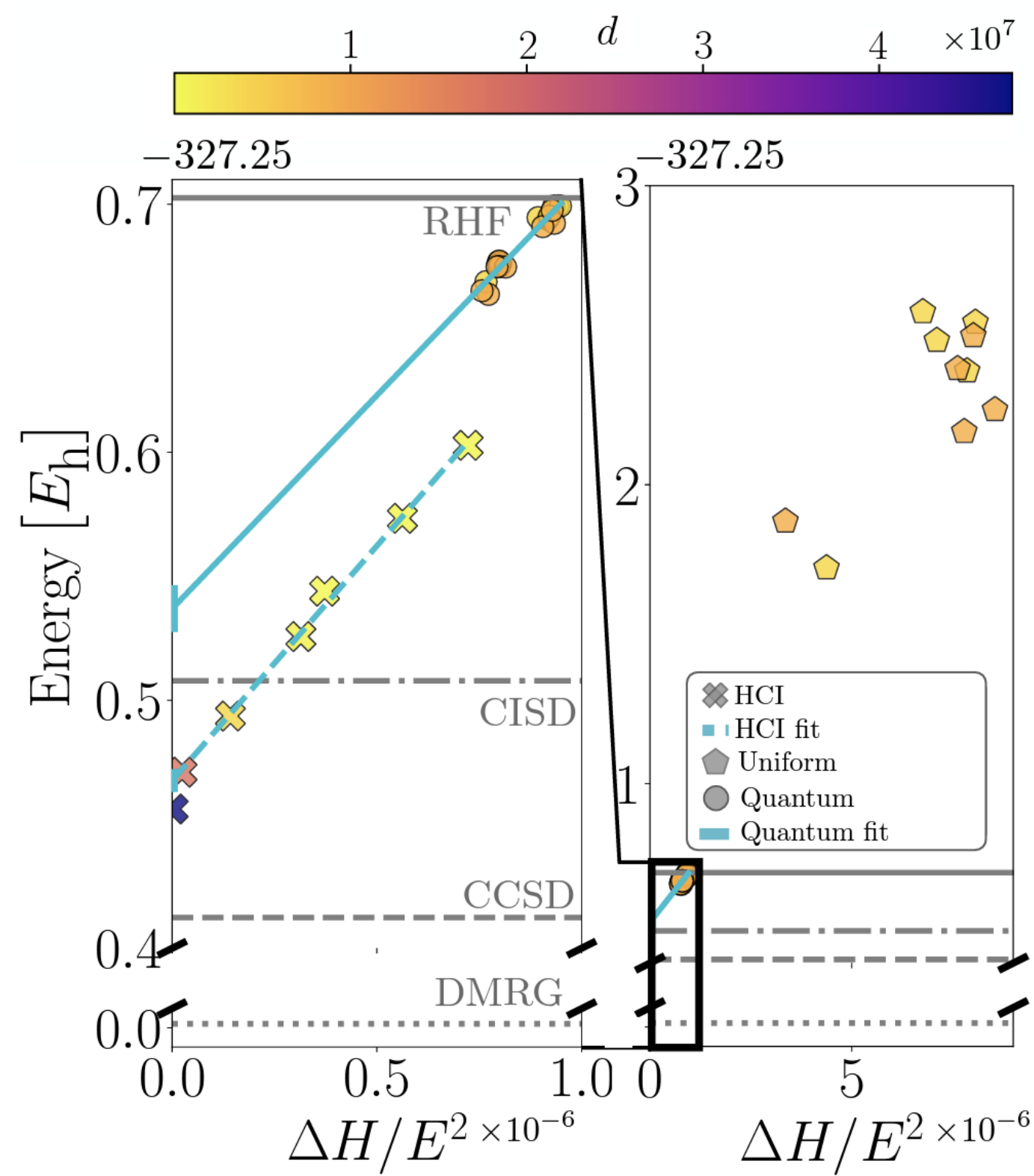
Project and diagonalize the Hamiltonian over each $\mathcal{S}(k)$, using iterative Davidson method (Eigenstate solver) on multiple classical nodes.
- Step 6

Resulting ground states are used to obtain a new occupancy
$$n_{p\sigma} = \frac{1}{K} \sum_{1 \leq k \leq K} \langle \psi^{(k)} | \hat{n}_{p\sigma} | \psi^{(k)} \rangle$$
and the minimum energy is the average over the different batches.
The configuration recovery loop is repeated until convergence.

SQD results: Pushing hardware capabilities

Chemistry Beyond Exact Solutions on a Quantum-Centric Supercomputer

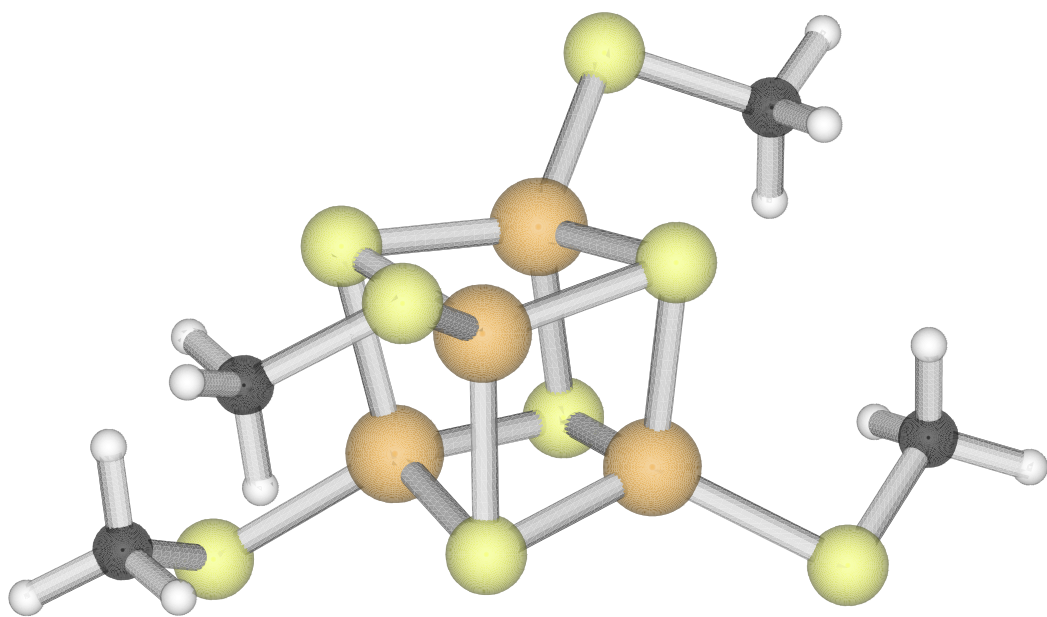
Javier Robledo-Moreno,^{1,*} Mario Motta,^{1,†} Holger Haas,¹ Ali Javadi-Abhari,¹ Petar Jurcevic,¹ William Kirby,² Simon Martiel,³ Kunal Sharma,¹ Sandeep Sharma,⁴ Tomonori Shirakawa,^{5,6,7} Iskandar Sitdikov,¹ Rong-Yang Sun,^{5,6,7} Kevin J. Sung,¹ Maika Takita,¹ Minh C. Tran,² Seiji Yunoki,^{5,6,7,8} and Antonio Mezzacapo^{1,‡}



The largest quantum circuit for molecular simulations:
active space Fe_4S_4

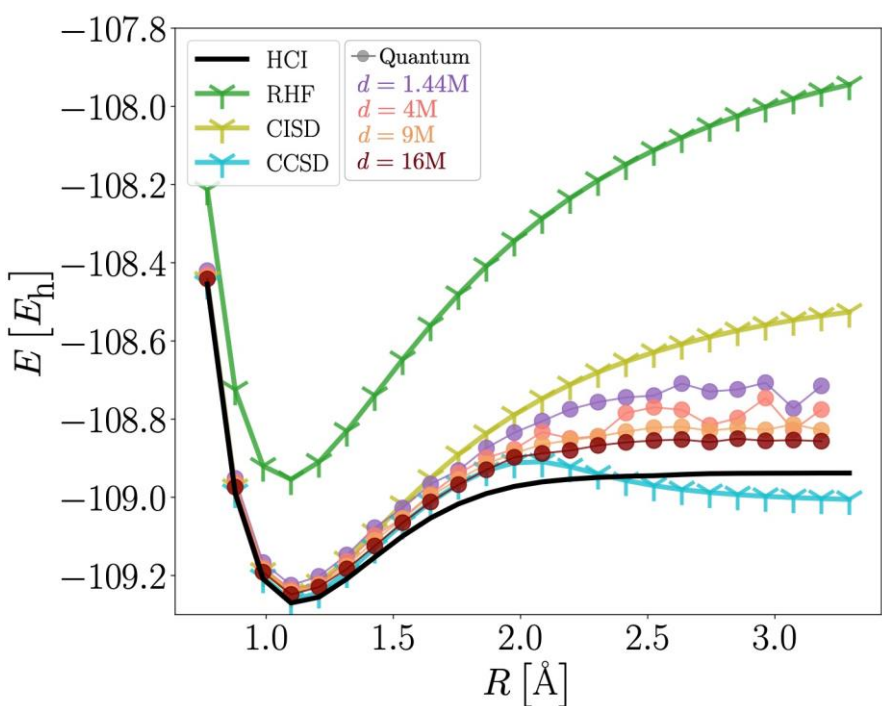
77 qubits
10570 single qubit gates
3590 two-qubit gates

Pre-fault-tolerant quantum data on a circuit of 10k gates
can provide a useful signal for chemistry



SQD results: Pushing hardware capabilities

Chemistry beyond exact solutions...

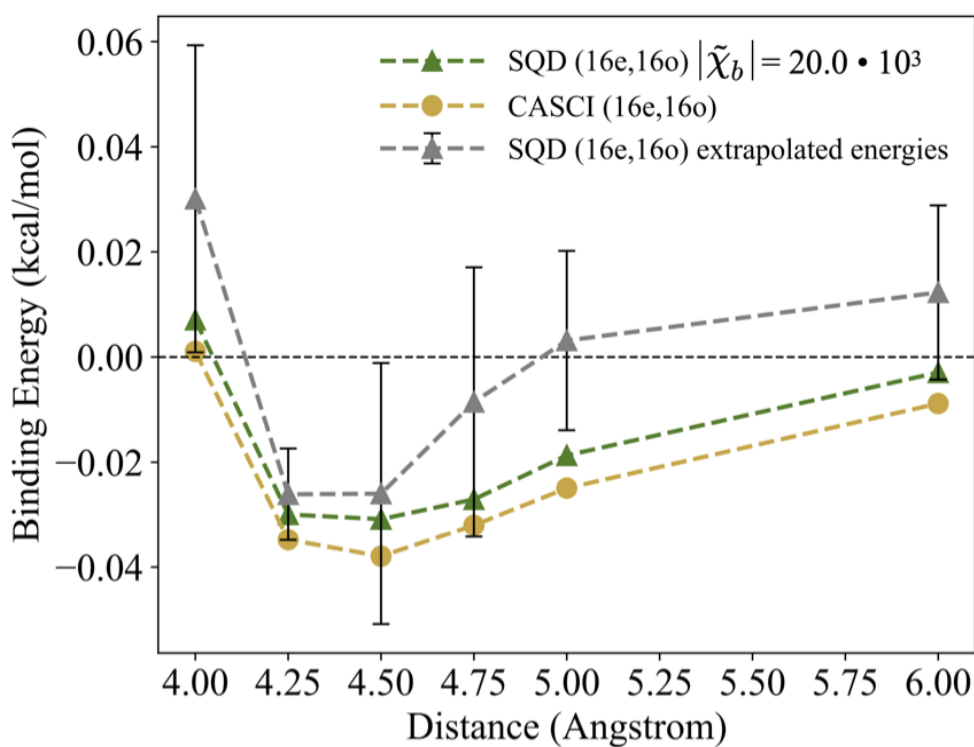


77 qubits
3500 2-qubit gates

RIKEN
IBM Quantum

arXiv:2405.05068

High-precision SQD applied to supramolecular interactions

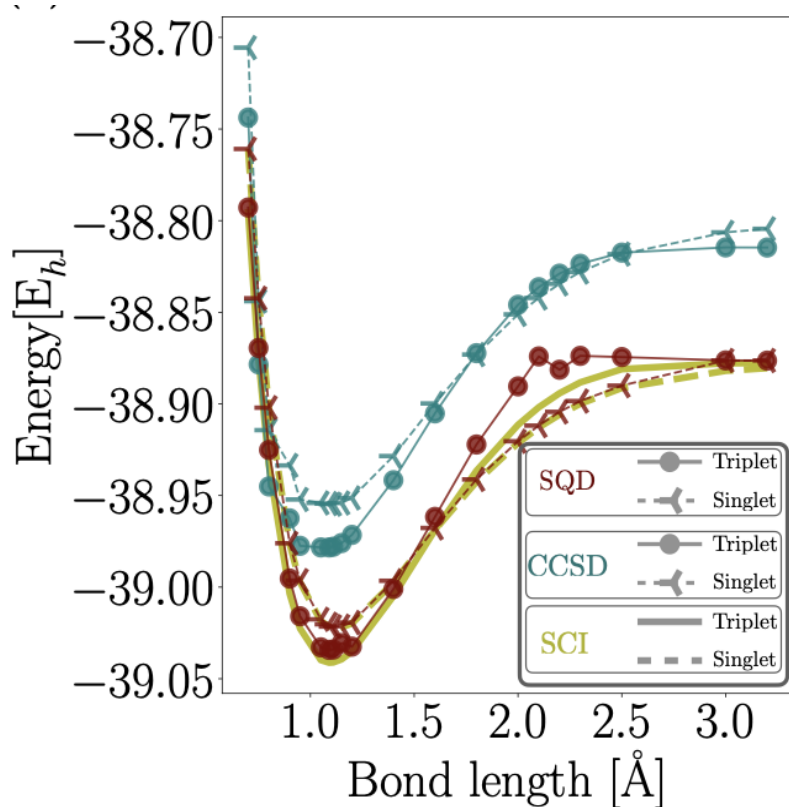


52 qubits
1600 2-qubit gates

Cleveland Clinic
IBM Quantum

arXiv:2410.09209

SQD applied to open-shell systems

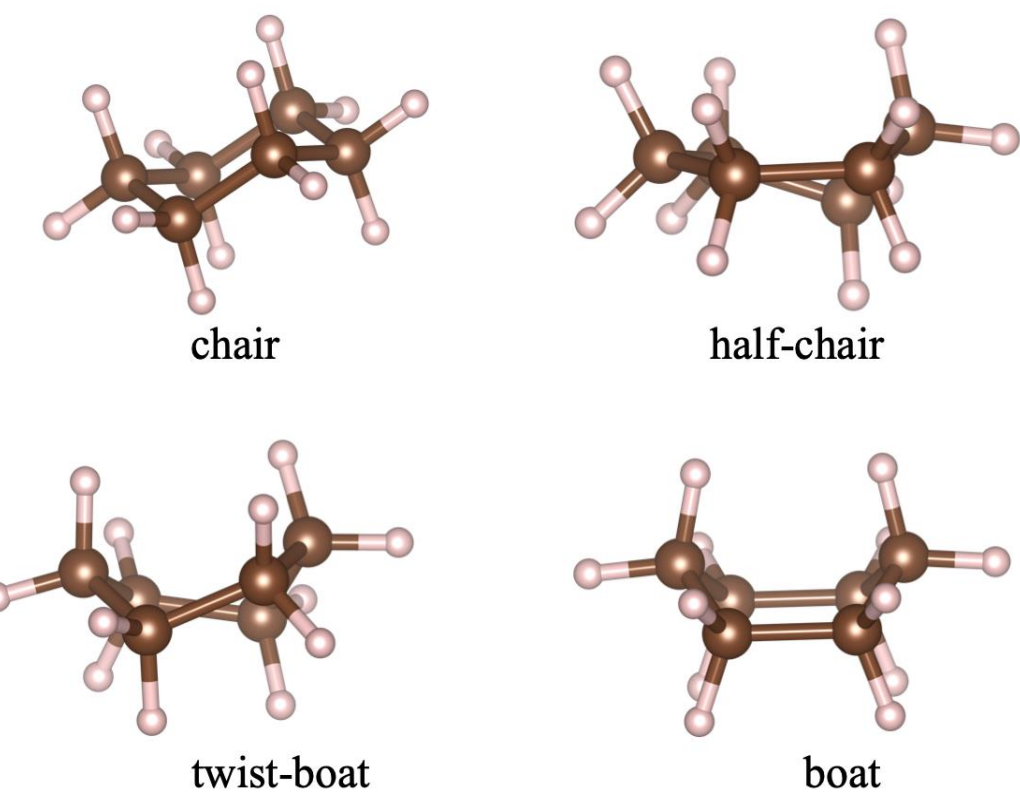


52 qubits
3500 2-qubit gates

Lockheed Martin
IBM Quantum

arXiv:2411.04827

DMET + SQD for organic chemistry

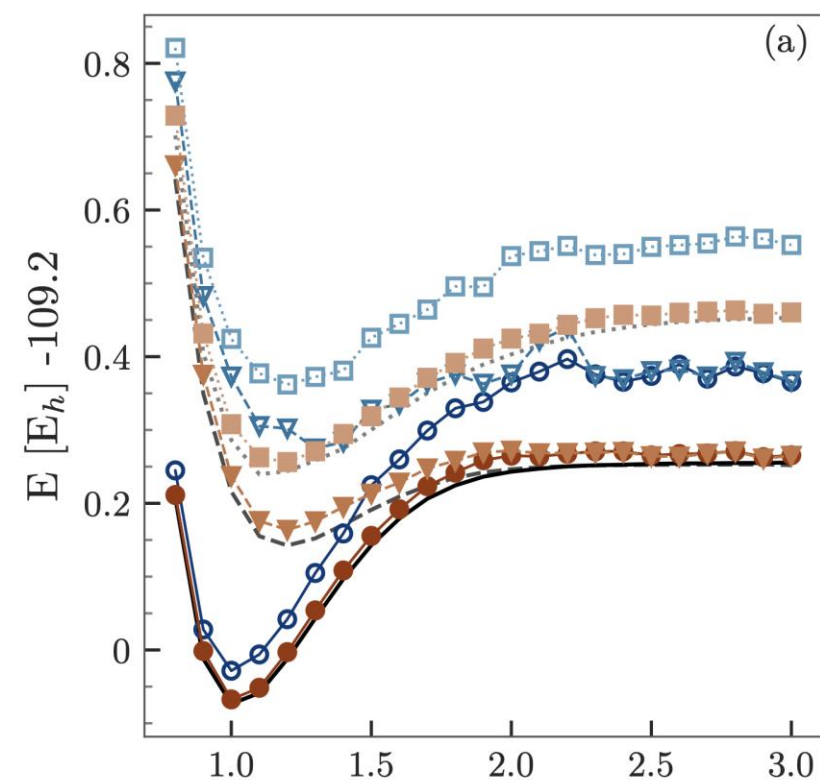


41 qubits
1500 2-qubit gates

Cleveland Clinic
IBM Quantum

arXiv:2411.09861

Ext-SQD: excited states

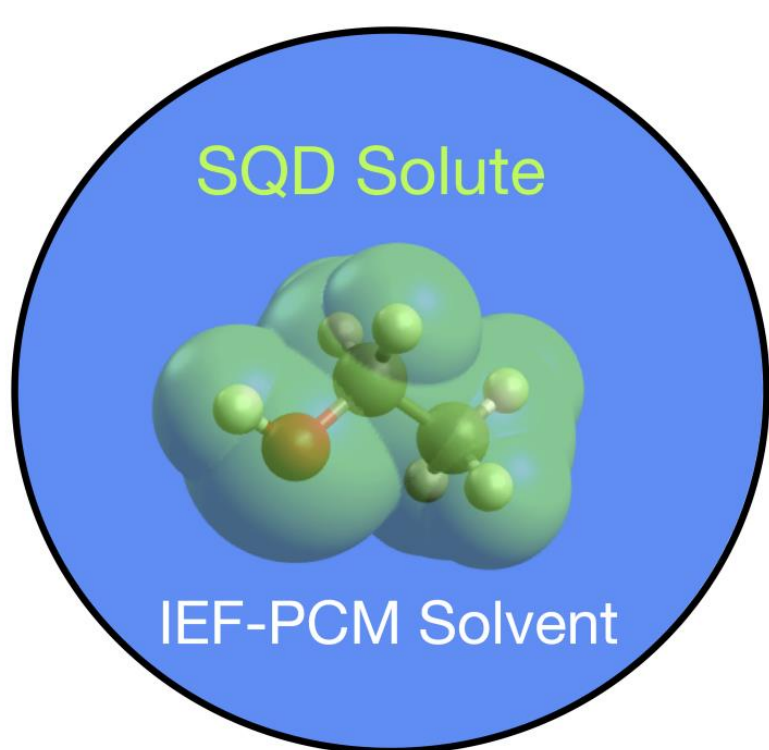


58 qubits
1792 2-qubit gates

IBM Quantum

arXiv:2411.00468

SQD + IEF-PCM for solute-solvent interactions



52 qubits
2422 2-qubit gates

Cleveland Clinic

arXiv:2502.10189

SQD (some) take aways

- Quantum-centric supercomputing enables realistic use cases beyond problems tailored to the device connectivity.

- Hybrid approach: QPU executes a few large quantum circuits and HPC processes large classical data.

- Processing of quantum data at the sample level avoids false positive solutions and provides certifiable advantage.

- The goal of the quantum circuit is not to exactly approximate the probability distribution underlying the ground state, but to sample from its support – or a subset of it - to give the lowest energy possible.

- There is value in trying the approach with different input circuits because any solution can be certified and ranked by quality.

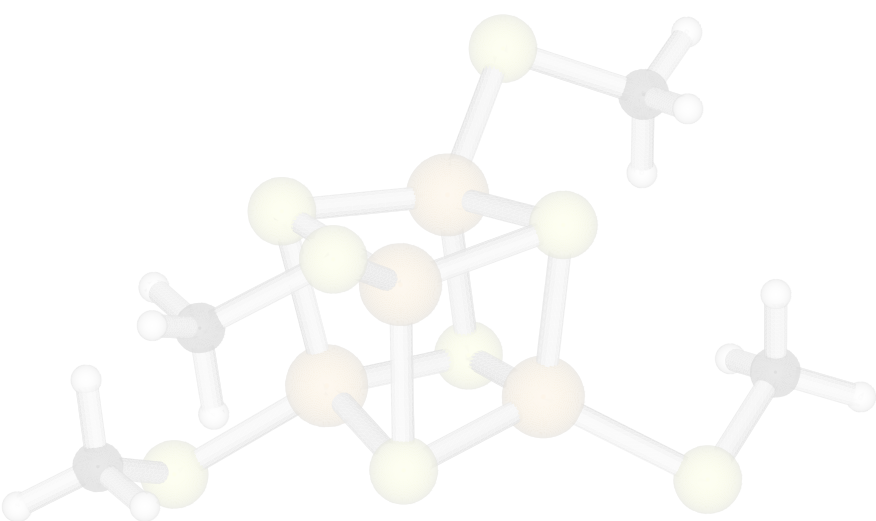
- The approximate solutions can be stored in classical memory, which permits further classical processing.

Quantum Diagonalization Methods

1. Sample-based Quantum Diagonalization (SQD) May 2024

Chemistry Beyond Exact Solutions on a Quantum-Centric Supercomputer

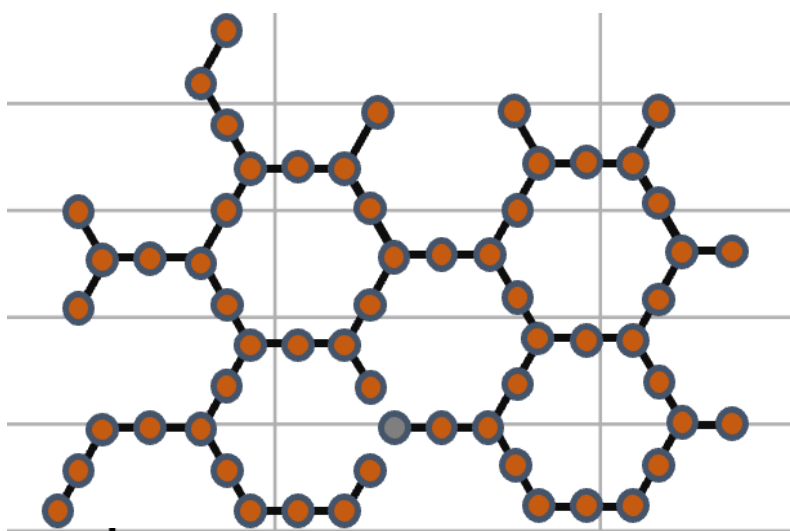
Javier Robledo-Moreno,^{1,*} Mario Motta,^{1,†} Holger Haas,¹ Ali Javadi-Abhari,¹ Petar Jurcevic,¹ William Kirby,² Simon Martiel,³ Kunal Sharma,¹ Sandeep Sharma,⁴ Tomonori Shirakawa,^{5,6,7} Iskandar Sitdikov,¹ Rong-Yang Sun,^{5,6,7} Kevin J. Sung,¹ Maika Takita,¹ Minh C. Tran,² Seiji Yunoki,^{5,6,7,8} and Antonio Mezzacapo^{1,‡}



2. Krylov Quantum Diagonalization (KQD) October 2024

Diagonalization of large many-body Hamiltonians on a quantum processor

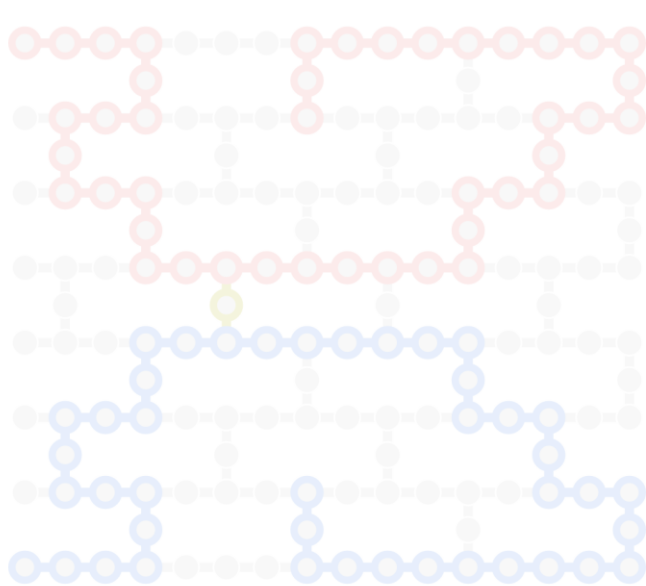
Nobuyuki Yoshioka*,^{1,†} Mirko Amico*,^{2,‡} William Kirby*,^{3,§} Petar Jurcevic,² Arkopal Dutt,³ Bryce Fuller,² Shelly Garion,⁴ Holger Haas,² Ikko Hamamura**,⁵ Alexander Ivrii,⁴ Ritajit Majumdar,⁶ Zlatko Minev,² Mario Motta,² Bibek Pokharel,⁷ Pedro Rivero,² Kunal Sharma,² Christopher J. Wood,² Ali Javadi-Abhari,² and Antonio Mezzacapo²



3. Sample-based Krylov Quantum Diagonalization (SKQD) January 2025

Sample-based Krylov Quantum Diagonalization

Jeffery Yu,^{1,2,3,*} Javier Robledo Moreno,^{1,†} Joseph Iosue,^{1,2,3} Luke Bertels,⁴ Daniel Claudino,⁴ Bryce Fuller,¹ Peter Groszkowski,⁵ Travis S Humble,⁶ Petar Jurcevic,¹ William Kirby,⁷ Thomas A. Maier,⁸ Mario Motta,¹ Bibek Pokharel,¹ Alireza Seif,¹ Amir Shehata,⁶ Kevin J. Sung,¹ Minh C. Tran,¹ Vinay Tripathi,¹ Antonio Mezzacapo,^{1,‡} and Kunal Sharma^{1,§}



Krylov Quantum Diagonalization (KQD)

Workflow:

- 1. Start with initial guess $|\psi_0\rangle$
- 2. Define Krylov vectors $V = [|\psi_0\rangle, U|\psi_0\rangle, U^2|\psi_0\rangle, \dots, U^{D-1}|\psi_0\rangle]$ for $U = e^{iHdt}$
- 3. $H_S =$ project H onto span V , the Krylov space
- 4. Diagonalize the H_S
- 5. Lowest eigenvalue of H_S , i.e., of $Hv = \lambda Sv$, approximates lowest eigenvalue of H

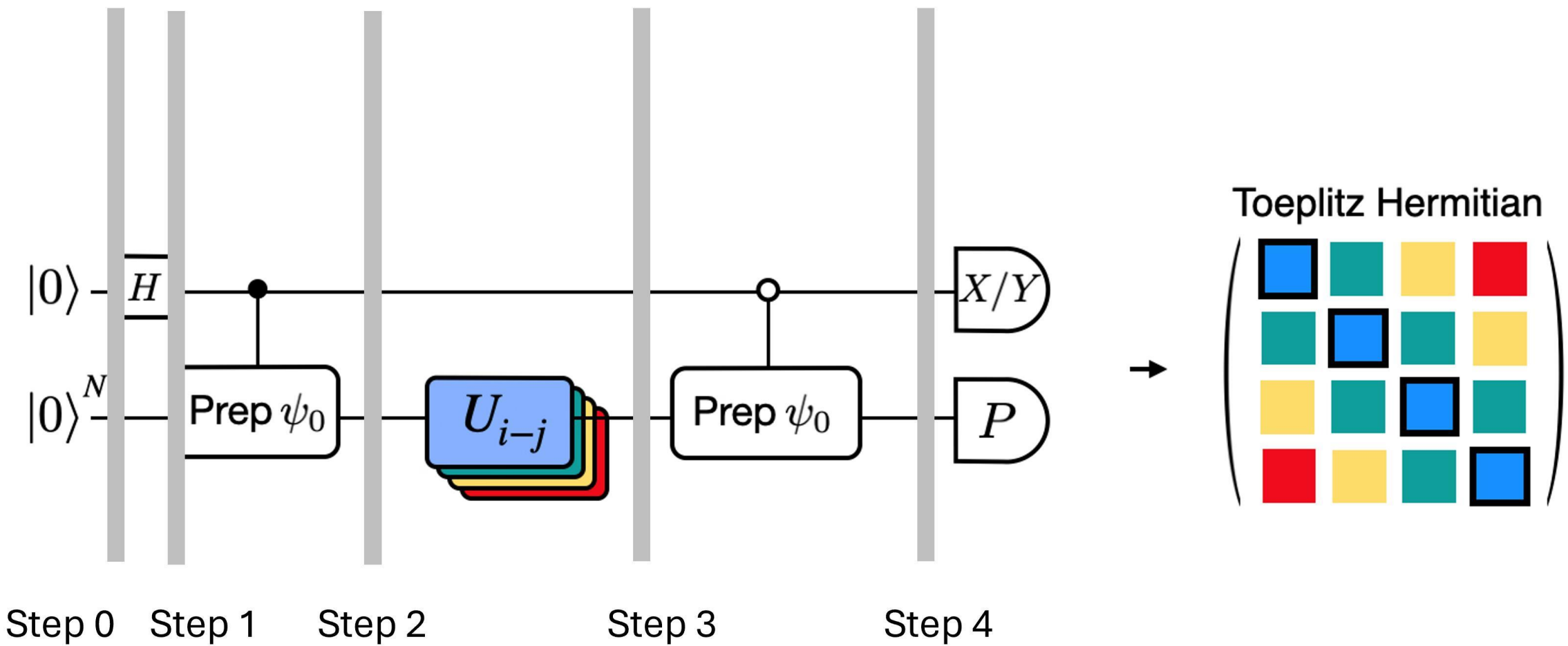
$V^\dagger$

H

V

 $=$

H_S

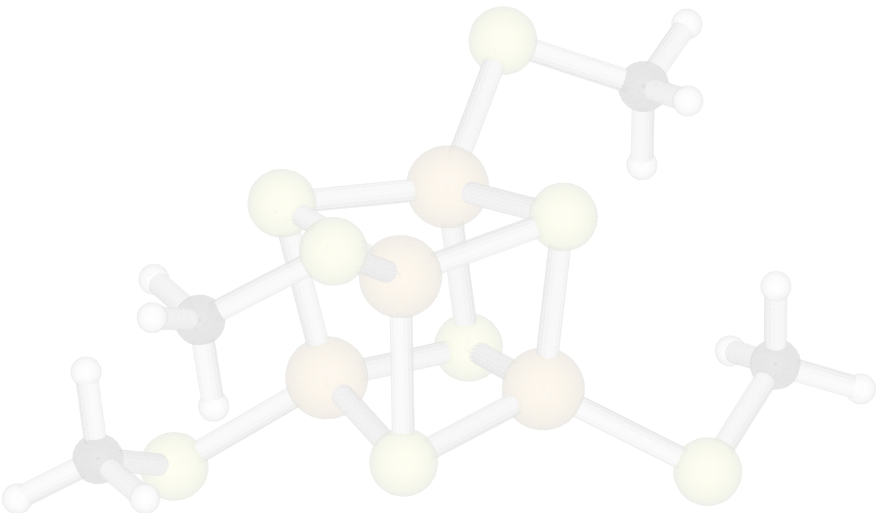


Quantum Diagonalization Methods

1. Sample-based Quantum Diagonalization (SQD) May 2024

Chemistry Beyond Exact Solutions on a Quantum-Centric Supercomputer

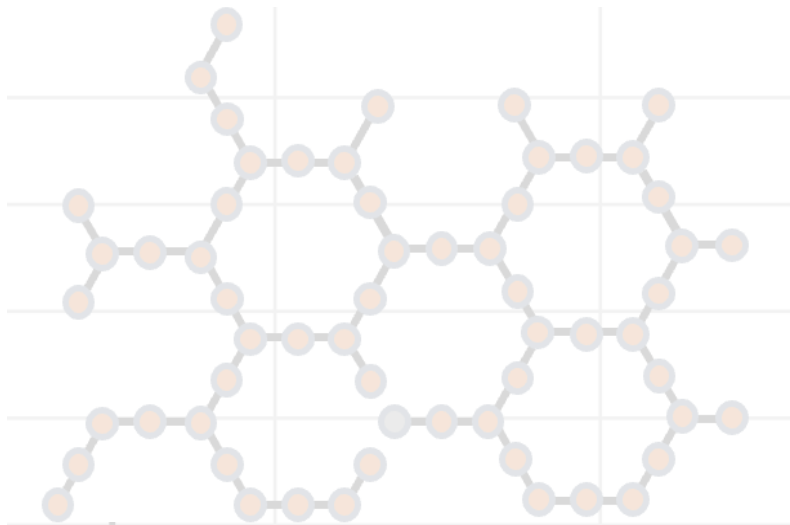
Javier Robledo-Moreno,^{1,*} Mario Motta,^{1,†} Holger Haas,¹ Ali Javadi-Abhari,¹ Petar Jurcevic,¹ William Kirby,² Simon Martiel,³ Kunal Sharma,¹ Sandeep Sharma,⁴ Tomonori Shirakawa,^{5,6,7} Iskandar Sitdikov,¹ Rong-Yang Sun,^{5,6,7} Kevin J. Sung,¹ Maika Takita,¹ Minh C. Tran,² Seiji Yunoki,^{5,6,7,8} and Antonio Mezzacapo^{1,‡}



2. Krylov Quantum Diagonalization (KQD) October 2024

Diagonalization of large many-body Hamiltonians on a quantum processor

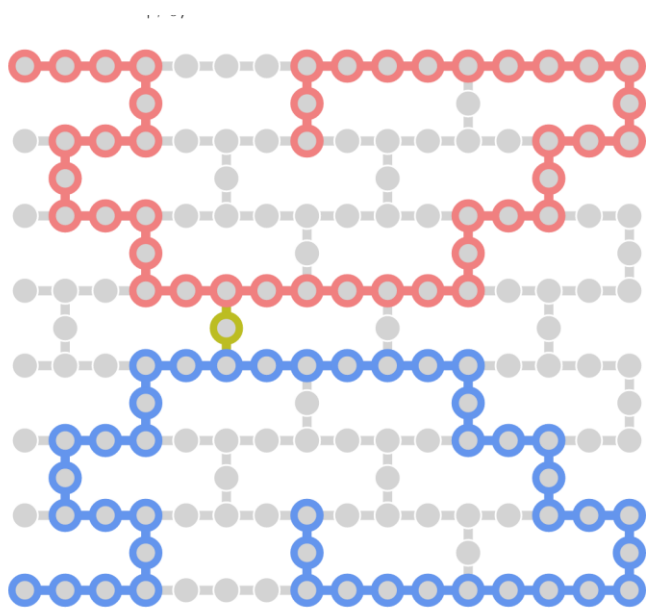
Nobuyuki Yoshioka*,^{1,†} Mirko Amico*,^{2,‡} William Kirby*,^{3,§} Petar Jurcevic,² Arkopal Dutt,³ Bryce Fuller,² Shelly Garion,⁴ Holger Haas,² Ikko Hamamura**,⁵ Alexander Ivrii,⁴ Ritajit Majumdar,⁶ Zlatko Minev,² Mario Motta,² Bibek Pokharel,⁷ Pedro Rivero,² Kunal Sharma,² Christopher J. Wood,² Ali Javadi-Abhari,² and Antonio Mezzacapo²



3. Sample-based Krylov Quantum Diagonalization (SKQD) January 2025

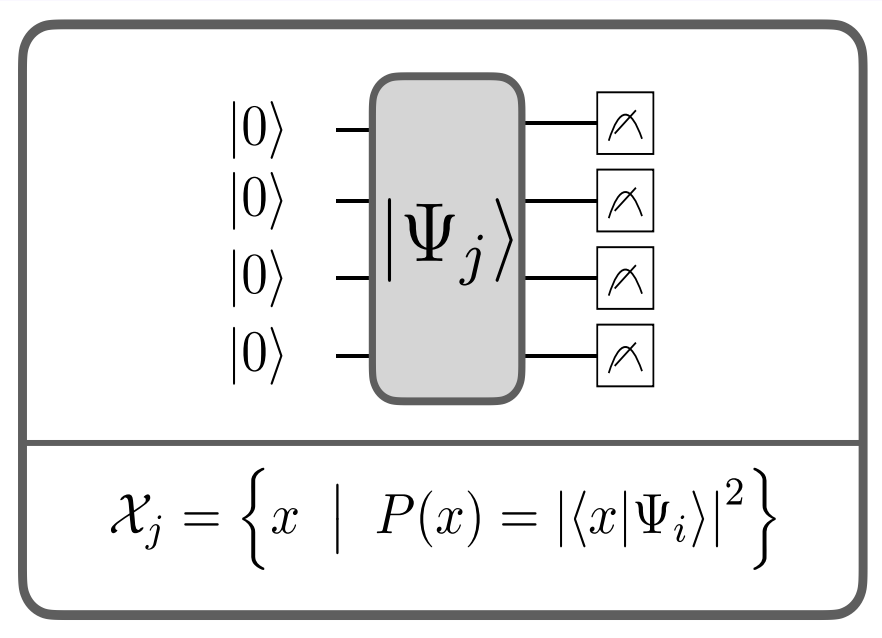
Sample-based Krylov Quantum Diagonalization

Jeffery Yu,^{1,2,3,*} Javier Robledo Moreno,^{1,†} Joseph Iosue,^{1,2,3} Luke Bertels,⁴ Daniel Claudino,⁴ Bryce Fuller,¹ Peter Groszkowski,⁵ Travis S Humble,⁶ Petar Jurcevic,¹ William Kirby,⁷ Thomas A. Maier,⁸ Mario Motta,¹ Bibek Pokharel,¹ Alireza Seif,¹ Amir Shehata,⁶ Kevin J. Sung,¹ Minh C. Tran,¹ Vinay Tripathi,¹ Antonio Mezzacapo,^{1,‡} and Kunal Sharma^{1,§}



Sample-based Krylov quantum diagonalization (SKQD)

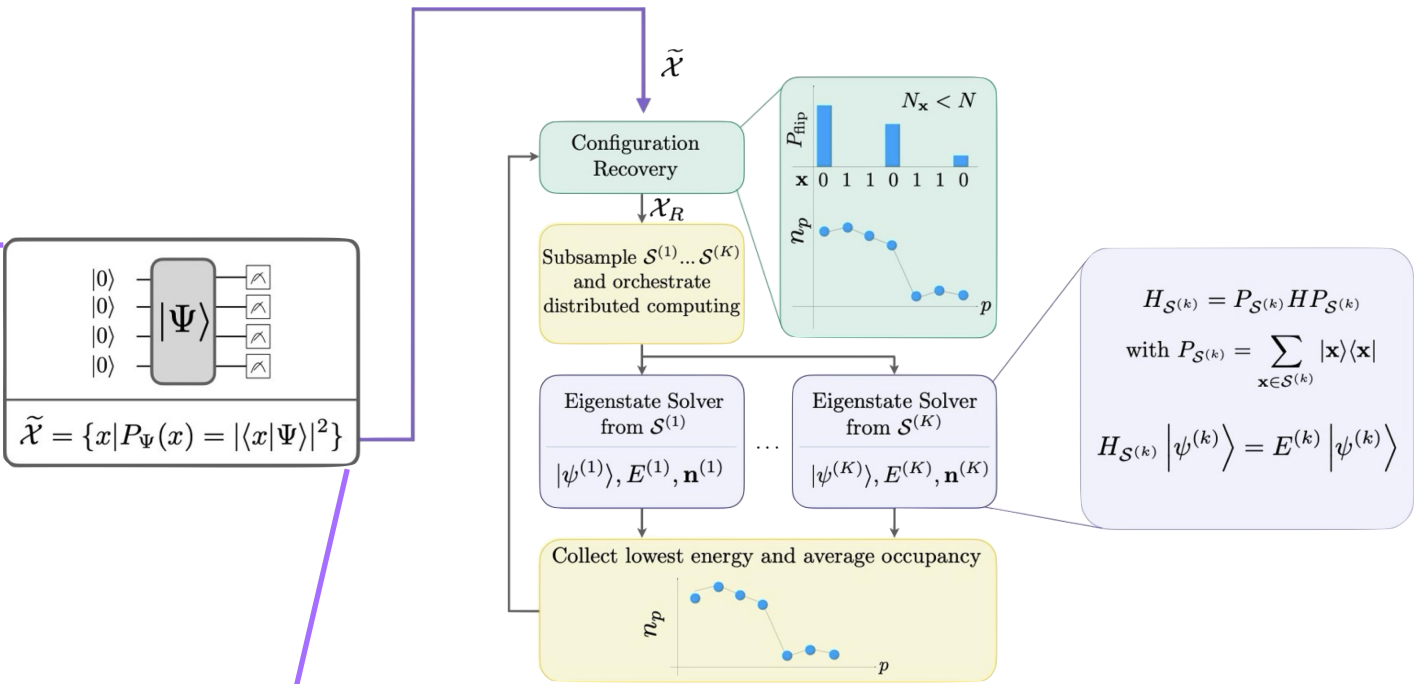
Sampler()



Simple states to prepare that have good overlap with ground states

$$|\Psi_j\rangle = |\phi_{\text{ref}}\rangle \exp(-ij\Delta t H) \quad \text{Krylov Vector}$$

Quantum devices are naturally suited for performing **time evolution circuits**.



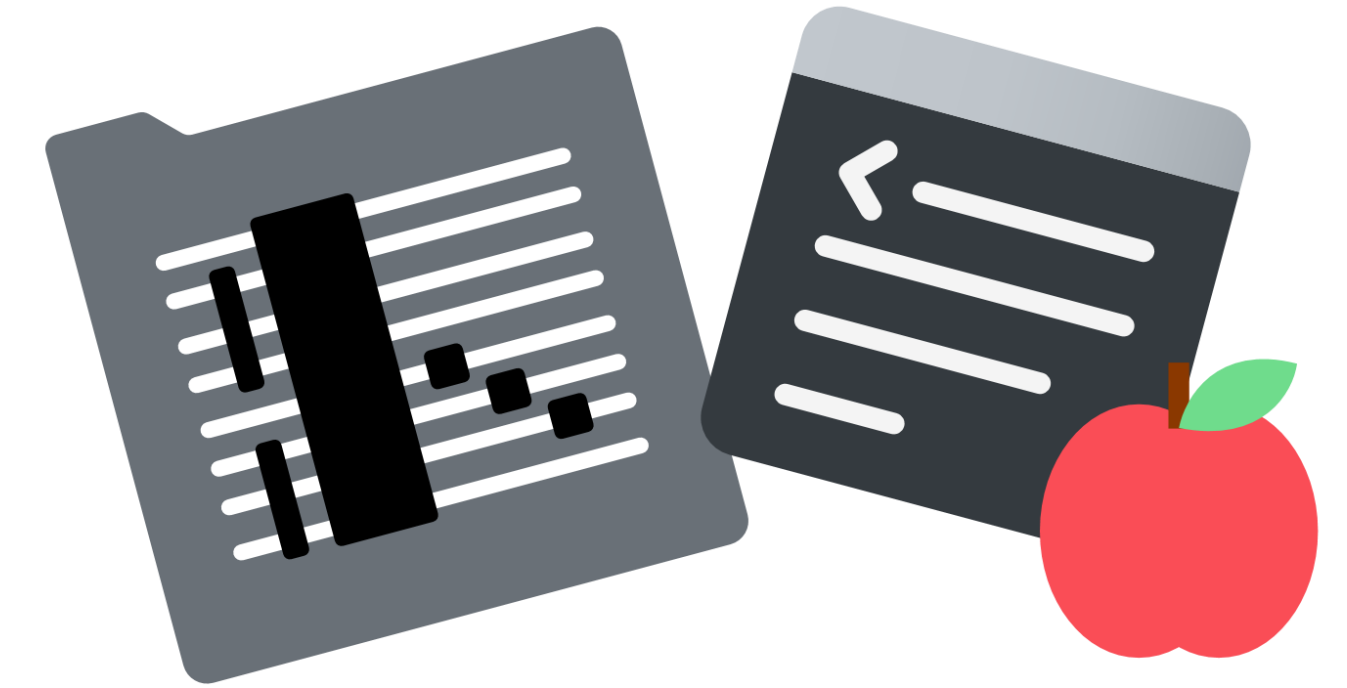
SQD vs SKQD

Commonalities

- Give **no false positives** for ground states.
- Can take advantage of **large-scale distributed classical computation**.
- Achieves complete **mitigation of quantum noise**, given enough running time.

Differences

- SKQD is **provably convergent** while SQD is not.
- **SKQD** doesn't require a variational **Ansatz** while SQD does (hardware efficient).
- **SKQD** uses crude approximations for time-evolution to get **low circuit depth**.



Further reading

Documentation:

<https://quantum.cloud.ibm.com/docs/en/guides/qiskit-addons-sqd>

Courses:

SQD Implementation:

<https://quantum.cloud.ibm.com/learning/en/courses/quantum-diagonalization-algorithms/sqd-implementation>

SKQD

<https://quantum.cloud.ibm.com/learning/en/courses/quantum-diagonalization-algorithms/skqd>

Tutorials:

SQD:

<https://quantum.cloud.ibm.com/docs/en/tutorials/sample-based-quantum-diagonalization>

KQD:

<https://quantum.cloud.ibm.com/learning/en/courses/quantum-diagonalization-algorithms/krylov>